

A Mathematical Optimisation Framework to Medium-Term Production Planning in the Metal Packaging Industry

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Master's Dissertation

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Master's in Industrial Engineering and Management

2026-06-25

Abstract

The metal packaging industry presents specific production planning challenges arising from the combination of highly customized products, multi-level bills of materials, and a sequential manufacturing process encompassing varnishing, lithography, stamping, and assembly. Each finished product is usually tied to a single customer, the active portfolio is large, and demand is both seasonal and uncertain. Coordinating material flows and intermediate inventories across successive production stages, while reconciling the competing objectives of high service levels and low operational costs, is a demanding task. Despite its complexity, this process is still frequently handled through fragmented and experience-driven decision-making, where forecasting, production planning, and procurement are treated in isolation. This separation often translates into excess inventory, underused capacity, or an inability to respond to customer demand on time.

This dissertation, developed in collaboration with LTPlabs for a Brazilian metal packaging manufacturer, addresses these challenges by proposing a medium-term production planning model. The model constitutes the central component of a broader four-module decision framework that also comprises demand forecasting, short-term production planning, and coil procurement, with outputs flowing systematically from one module to the next. The medium-term model is formulated as a Mixed-Integer Linear Program and generates weekly production plans over a rolling 32-week horizon. It determines the production quantities of each product per machine and week, the consumption of components across the multi-level product structure, the procurement of steel coils, and the resulting inventory trajectories, while respecting capacity, shift availability, multi-stage routing, and parent-component relationships defined by the Bill of Materials.

A central feature of the formulation is a layered soft-constraint structure that governs the fulfilment of confirmed orders, forecasted demand, and inventory targets such as safety and cycle stock. Rather than imposing these requirements as hard constraints, the model penalizes their violation in the objective function, allowing the relative priority assigned to each to be reflected in the optimisation and preserving feasibility under tight capacity. This design enables planners to balance customer commitments against the cost of holding inventory and performing setups, producing plans that anticipate seasonal demand peaks by building stock in advance while smoothing resource utilization across the planning horizon.

The model was implemented in Python with the Pyomo library and solved with Gurobi, and was applied to real data from the company over a planning horizon of eight months. It generates weekly production plans that specify, for each product, the quantities to produce per machine, the components consumed across the multi-level product structure, the coils to procure, and the resulting inventory levels throughout the planning horizon. By integrating these decisions within a single optimisation framework, the model provides the company with a transparent and structured planning tool that replaces fragmented, experience-driven practices and supports more informed and coherent decisions across the supply chain.

Resumo

A indústria de embalagens metálicas opera num conjunto de condições que tornam o planeamento da produção particularmente desafiante: produtos altamente personalizados, estruturas de produto (*bill of materials*) multinível e um processo de fabrico sequencial que abrange envernizamento, litografia, estampagem e montagem. Cada produto final está normalmente associado a um único cliente, o portefólio ativo é extenso e a procura é simultaneamente sazonal e incerta. Coordenar os fluxos de materiais e os inventários intermédios ao longo das sucessivas etapas de produção, conciliando os objetivos concorrentes de elevados níveis de serviço e baixos custos operacionais, constitui uma tarefa exigente. Apesar da sua complexidade, este processo continua a ser frequentemente gerido através de decisões fragmentadas e baseadas na experiência, em que a previsão da procura, o planeamento da produção e o aprovisionamento são tratados de forma isolada. Esta separação traduz-se frequentemente em excesso de inventário, subutilização da capacidade produtiva ou incapacidade de responder atempadamente à procura dos clientes.

A presente dissertação, desenvolvida em colaboração com a LTPlabs para um fabricante brasileiro de embalagens metálicas, aborda estes desafios através da proposta de um modelo de planeamento da produção de médio prazo. O modelo constitui o componente central de um quadro de decisão mais abrangente, composto por quatro módulos, que inclui também a previsão da procura, o planeamento da produção de curto prazo e o aprovisionamento de bobinas de aço, com os resultados de cada módulo a alimentarem sistematicamente o seguinte. O modelo de médio prazo é formulado como um *Mixed-Integer Linear Program* e gera planos semanais de produção ao longo de um horizonte móvel de 32 semanas. Determina as quantidades a produzir de cada produto por máquina e por semana, o consumo de componentes ao longo da estrutura multinível dos produtos, a aquisição de bobinas de aço e as trajetórias resultantes dos inventários, respeitando simultaneamente restrições de capacidade, disponibilidade de turnos, roteiros produtivos multietapas e relações entre produtos-pai e componentes definidas na estrutura multinível dos produtos.

Uma característica central da formulação é a estrutura hierarquizada de restrições flexíveis (*soft constraints*) que rege o cumprimento de encomendas confirmadas, da procura prevista e de objetivos de inventário, como o stock de segurança e o stock de ciclo. Em vez de impor estes requisitos como restrições rígidas, o modelo penaliza o seu incumprimento na função objetivo, permitindo que a prioridade relativa atribuída a cada um seja refletida no processo de otimização e preservando a viabilidade da solução em cenários de capacidade limitada. Esta abordagem permite aos planeadores equilibrar os compromissos com os clientes face aos custos de posse de inventário e de realização de setups, gerando planos que antecipam picos sazonais de procura através da constituição antecipada de stock, ao mesmo tempo que promovem uma utilização mais uniforme dos recursos ao longo do horizonte de planeamento.

O modelo foi implementado em Python, utilizando a biblioteca Pyomo, resolvido com o Gurobi e aplicado a dados reais da empresa. Os planos semanais que produz indicam, para cada produto, as quantidades a fabricar por máquina, os componentes consumidos ao longo da estrutura multinível, as quantidades de aço a adquirir e os níveis de inventário ao longo do horizonte. Ao reunir todas

estas decisões numa única estrutura de otimização, o modelo dota a empresa de uma ferramenta de planeamento transparente e estruturada, capaz de substituir as práticas fragmentadas e assentes na experiência, e de apoiar decisões mais informadas e coerentes em toda a cadeia de abastecimento.

Agradecimentos

Este documento assinala o culminar de um período de cinco anos, marcado por inúmeras aprendizagens, momentos felizes e um crescimento contínuo. Por isso, começo por agradecer a todos com quem me cruzei ao longo desta etapa, e que contribuíram, direta ou indiretamente, para que tão bem sucedida tenha sido.

Estes últimos meses foram intensos — foi o período em que mais aprendi desde que escolhi o rumo para o meu futuro, há cinco anos. Por isso, agradeço à LTPlabs, pela confiança no meu trabalho e às pessoas que, nesse ambiente, me apoiaram ao longo desta maratona: agradeço à Catarina, minha orientadora na empresa, pelo suporte contínuo e recheado de ensinamentos, e ao Bruno e ao Rodrigo, com quem tive o prazer de trabalhar diariamente e de tanto aprender.

Agradeço muito à Professora Henriqueta Nóvoa, que foi incansável no processo de orientação, pela disponibilidade e acompanhamento louváveis — não só pelo apoio na redação do documento, mas principalmente pela motivação que me foi transmitindo, tendo tornado este caminho mais simples.

Por último, agradeço às pessoas que têm sido presença constante nas várias etapas da minha vida por tudo o que me têm transmitido. Agradeço à minha família — à minha mãe, ao meu pai e à minha irmã, que me apoiam, de forma constante, em todos os momentos, transformando os mais desafiantes em demonstrações de resiliência e celebrando sempre os mais alegres. Agradeço às minhas tias — em especial à Tia Carla, pelo apoio direto e indireto, que foi essencial na superação dos momentos difíceis que enfrentei. À minha avó, pela forma alegre como vive a vida, e como assim me ensina a viver, com esperança. Agradeço aos meus amigos por todos os conselhos, risos e desabafos. À Doutora Sílvia e ao Doutor Fernando, um agradecimento especial, pelo acompanhamento ao longo destes meses.

Em suma, agradeço a todas as pessoas que têm feito e continuarão, certamente, a fazer da minha vida mais bonita, e que me deixam ansioso pelas próximas etapas que serão recheadas de aprendizagens, porque só assim sei viver.

Um sincero e profundo obrigado,
Mateus Carmesim Marques

"In the depth of winter, I finally learned that within me there lay an invincible summer."

Albert Camus

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Acronyms and Symbols

BOM	Bill of Materials
LP	Linear Programming
MILP	Mixed-Integer Linear Programming
MPS	Master Production Scheduling
MRP	Material Requirements Planning
MTO	Make-to-Order
MTS	Make-to-Stock
OEE	Overall Equipment Effectiveness

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Chapter 1

Introduction

In today's world, marked by a VUCA context — Volatility, Uncertainty, Complexity, and Ambiguity — and increasing competitiveness in global markets, it is imperative to support decision-making processes with objective and data-driven tools. These tools are informed by the environment in which companies operate and aim to guide decision-makers towards the most appropriate course of action.

Analytics serve as a foundation for informed decision-making and can be classified into three categories: descriptive, predictive or prescriptive. Descriptive analytics focuses on understanding what has happened in the past, typically through data mining techniques. Predictive analytics seeks to anticipate future events by applying statistical methods and forecasting models. Finally, prescriptive analytics builds upon these insights, using optimisation algorithms to recommend actions that lead to the best possible outcomes (Roy et al., 2022).

Decision-making within organizations can also be structured across three hierarchical levels: strategic, tactical, and operational. Strategic planning deals with long-term decisions — typically considering a two- to five-year planning horizon —, such as facility layout and resource capacity planning. Tactical planning focuses on medium-term decisions, including inventory management. Operational planning, in turn, deals with short-term decisions, such as production scheduling and sequencing (Torabi et al., 2010).

In industrial companies, production, the process of transforming raw materials into finished products, lies at the core of business operations, making effective planning essential for overall performance. Production planning, therefore, leverages the vast amount of data that is available across organizations and supply chains to make the best decision on acquisition, allocation and utilization of production resources.

Production planning is inherently cross-functional, involving multiple departments such as production, procurement, and marketing, as well as external stakeholders across the supply chain, from suppliers to customers. Its impact extends beyond internal efficiency and cost minimization, contributing also to improved service levels by ensuring that customer demand is met in a timely and reliable manner.

While these concepts provide a strong theoretical foundation, their practical implementation

in real industrial environments remains challenging. Increasing system complexity, uncertainty in demand, and interdependencies across the supply chain make it difficult to rely solely on human intuition for decision-making.

However, there are still many companies that rely on empirical knowledge and human intuition to support decision-making processes. This approach presents significant risks, as it depends heavily on individual experience and is vulnerable to workforce turnover. Moreover, given the increasing complexity of production systems and the large number of feasible decision alternatives, relying solely on intuition becomes inefficient and suboptimal.

In this context, the adoption of optimisation models emerges as a key enabler for more robust and informed decision-making, allowing companies to improve service levels while reducing operational costs.

The motivation for this project arises from the opportunity to optimise production planning, a critical process in industrial environments. Medium-term production planning combines prescriptive analytics with tactical decision-making. In this context, production planning must balance the trade-off between inventory holding and setup costs, a key aspect in defining efficient production policies, while also contributing to higher service levels and customer satisfaction.

1.1 The Underlying Project

The project on which this dissertation is based was developed at LTPlabs, an analytics-driven management consultancy that combines advanced analytics and artificial intelligence with strong business expertise, for a Brazilian company in the metal packaging industry.

This work is part of a broader initiative aimed at enhancing decision-making processes across the entire supply chain, namely in demand planning, production planning, and procurement. In this context, the development of analytical and optimisation-based tools plays a key role in enabling more efficient and integrated planning decisions.

1.1.1 Industrial Context

This dissertation is situated within the metal packaging industry, which is characterized by complex production processes, multi-level product structures, and strong interdependencies between operations. Production typically starts from raw materials such as steel coils, which are transformed into intermediate components and later assembled into final products.

This multi-stage production system introduces additional challenges, namely in coordinating material flows, managing intermediate inventories, and synchronizing production across different work centers. The significant differences in processing times across operations can lead to bottlenecks and intermediate stock accumulation, further complicating the planning process. Furthermore, variability in demand and raw material supply adds uncertainty to planning decisions, increasing the need for robust and integrated approaches. The prevalence of small production lots also results in frequent setup operations, which, in this context, can last several hours, significantly impacting

available productive capacity. Addressing this issue is one of the key contributions of the developed model, which consolidates the production of each item into a single weekly run, thereby reducing the number of setups and improving overall efficiency.

1.1.2 Problem Statement

In practice, many industrial companies still rely on fragmented planning processes, where demand forecasting, production planning, and procurement decisions are handled separately. This lack of integration often leads to suboptimal decisions, such as excess inventory, capacity underutilization, or inability to meet customer demand.

Additionally, planners face multiple trade-offs, including balancing service levels with operational costs, managing inventory versus production efficiency, and allocating limited resources across competing products. These challenges are particularly relevant in medium-term planning, where decisions must anticipate future conditions while remaining feasible at the operational level.

1.1.3 Mission and Objectives

In the context studied, metal packaging is primarily used for food and chemical products, such as paints.

The metal packaging industry is particularly exposed to raw material price fluctuations and sourcing variability, which are recognized sources of supply chain risk (Tang, 2006). As such, effective coordination of the supply chain is essential, particularly in what concerns procurement decisions, which should be supported by analytical tools.

Given this context, the main objective of the project is to develop analytical tools capable of supporting integrated decision-making across the supply chain.

To address these challenges, the project followed a holistic and integrated approach to demand, production and procurement planning.

1.1.4 Project Plan

To tackle the challenges described above, the project structured its response around four analytical modules that together span the full decision-making chain — from raw material procurement to demand estimation and production execution. These modules are not independent: the outputs of each stage systematically feed subsequent ones, ensuring coherence across planning levels. Figure 1.1 illustrates this architecture and the relationships between modules.

The first module, A, operates over an eight-month horizon at a weekly frequency. Using time-series forecasting techniques, it estimates future demand for each finished product (FP) based on historical customer data, producing quantity estimates, demand variability, and safety stock levels that serve as the primary input to Module B.

Medium-Term Production Planning is the central focus of this dissertation (Module B). Over the same eight-month horizon, it generates the production plan across machines, while respecting

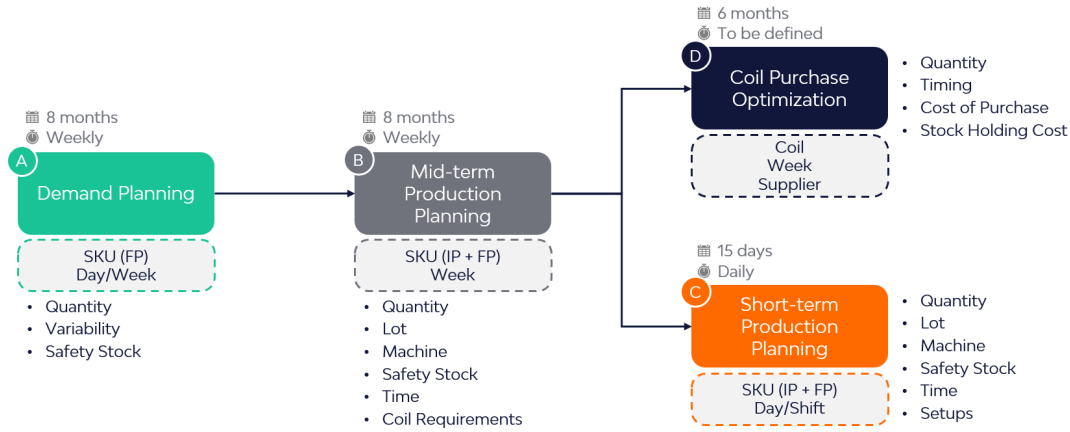


Figure 1.1: Project plan structure (IP: Intermediate Product; FP: Finished Product)

capacity constraints, inventory levels, and bill-of-materials structures. Its outputs also include coil consumption requirements and the necessary steel arrival schedules, which drive Module D.

Module C translates the medium-term plan into detailed, executable decisions over a 15-day horizon, updated daily, refining production allocation by machine, lot, and shift.

Steel Coil Purchase Optimisation (D) operates over a six-month horizon, using the raw material requirements generated by Module B to recommend optimal purchasing decisions (quantities, timing, and supplier selection), balancing order costs against tied-up inventory costs.

This data-driven, multi-level architecture replaces the fragmented, intuition-driven processes described previously with a structured decision-support system, ensuring that procurement, production, and demand decisions are no longer made in isolation but as part of a single, coherent planning process.

1.2 Scope of the Project

Although the project encompasses multiple analytical modules, this dissertation focuses specifically on the medium-term production planning component. This module is responsible for determining what to produce, in what quantities, and on which resources, on a weekly basis, over a planning horizon of approximately eight months.

Beyond defining a feasible production plan, the model delivers a comprehensive set of decision-support outputs that are particularly valuable in a volatile industrial environment, where customer orders arrive continuously and delivery commitments must be made with confidence. Specifically, the model:

- plans weekly production quantities for 32 weeks, determining how much of each product to produce in each period to satisfy demand, while accommodating demand uncertainty, through the maintenance of two stock components: safety stock, a buffer held to absorb

demand variability and prevent stockouts, and cycle stock, the inventory built up to cover demand until the next production run;

- provides a forward-looking view of production line capacity utilisation over the entire planning horizon, enabling planners to assess feasibility and support the definition of realistic delivery dates to customers;
- determines steel procurement requirements derived from planned production, ensuring raw material needs are anticipated well in advance;
- tracks inventory levels at every stage of the Bill of Materials — both intermediate and finished goods — over the full horizon, ensuring that warehouse occupancy remains within healthy bounds, while supporting uniform and consistent stock policies;
- offers a flexible and configurable framework in which planners can adjust the model's priorities and parameters to reflect the company's strategic objectives, such as assigning greater weight to Overall Equipment Effectiveness (OEE) or other Key Performance Indicators (KPIs).

By synthesising all of this information into a single coherent plan, the model acts as an integrated decision-support tool that brings structure and analytical rigour to a planning process currently driven by managerial intuition. It is particularly well-suited to the company's operational reality, where the high frequency of incoming customer orders demands a dynamic planning tool capable of providing a reliable and up-to-date view of production capacity. The model's outputs are presented through a web application (WebApp), which serves as the interface between the optimisation engine and the company's decision-makers. Through this platform, planners can analyse and evaluate the proposed production plan, approving or adjusting it as needed. The WebApp also allows key parameters to be configured directly — such as costs, penalizations, weights, lead times, and working days by shift — ensuring that the tool remains flexible and adaptable to the company's evolving operational needs.

The main contribution of this dissertation lies in the development of a mathematical optimisation model tailored to a real industrial context, capable of supporting tactical decision-making in medium-term production planning. In addition, the work contributes by integrating multiple supply chain dimensions — production, inventory, and procurement — incorporating realistic operational constraints, and providing a scalable framework applicable to similar industrial environments.

The schedule for the project is presented in Figure 1.2, detailing the allocation of each module over the full project horizon.

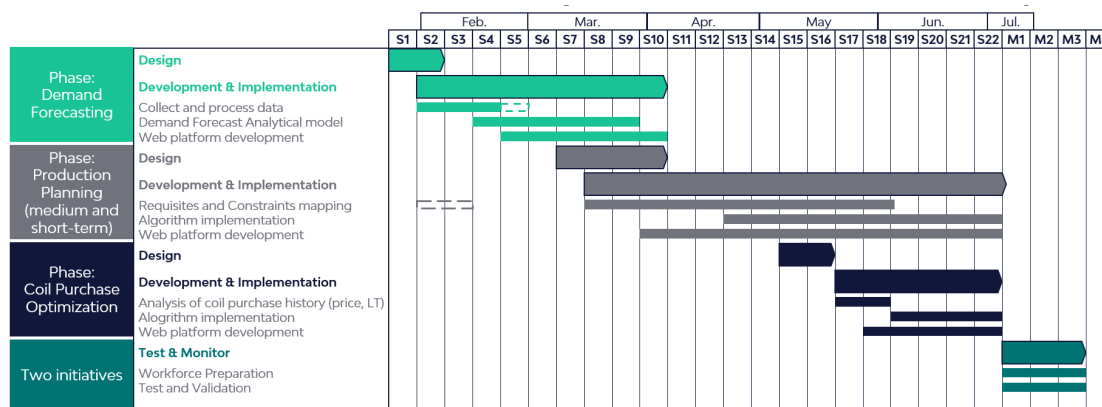


Figure 1.2: Schedule of the project

This project exhibits an inherent waterfall-like structure, as each module depends on the completion of the previous one. This dependency is especially evident in the medium-term production planning module, which uses inputs from demand forecasting and, in turn, provides outputs for the other two modules. The production planning modules required the greatest development effort, as they were the most complex and central components of the system.

1.3 Dissertation Structure

This document is organized into six chapters. Chapter 1 introduces the dissertation, providing industry context, stating the problem, and presenting the objectives, along with a brief overview of the approach followed. Chapter 2 establishes the theoretical foundations for the study, reviewing production planning with emphasis on its tactical level, divided into capacity planning and master production scheduling. This chapter also covers Materials Requirement Planning and closes with a discussion of solving approaches for optimisation models, with particular focus on Linear Programming. Chapter 3 details the problem under study, providing further industrial and planning context. The problem is then divided into two fronts, production process and data preparation, each explored in detail, and supported by an exploratory data analysis (EDA). Chapter 4 presents the methodology adopted to address the problem, focusing on the mathematical formulation of the optimisation model developed. Chapter 5 presents the results of the study, examining how the model outputs are presented to decision-makers and their impact on the company. Finally, Chapter 6 closes the document by summarizing the main conclusions and outlining potential future work.

Chapter 2

Theoretical Background

The objective of this chapter is to provide a theoretical foundation for the topics addressed throughout this dissertation. In order to achieve that, prevalent theories and studies on production planning subjects are explored in the next sections.

2.1 Production Planning

With increasing competition in global markets, effective planning has become progressively more important. Production planning is no exception — as production represents the core operation of industrial companies, its proper management is critical to overall performance.

Production planning is described by Pochet and Wolsey (2006) as the process of planning manufacturing activities to transform raw materials into finished products in order to meet customer demand efficiently. It is one of the most important decision-making processes within manufacturing systems and has gained increasing attention in recent decades (Demartini et al., 2021).

Although a production process begins with raw materials and ends with finished products, supply chain integration is crucial and must be considered within production planning. A supply chain can be defined as a network of organizations involved in the upstream and downstream flows of materials, information, and products, from raw material suppliers to end consumers (Mentzer et al., 2001). Effective production planning supports supply chain management by coordinating material requirements and product availability — so, production planning impacts both the upper and the lower sides of a supply chain, from procurement to distribution (Thomas and Griffin, 1996).

The classification of planning tasks can be, according to Anthony (1965), divided into the following three levels, with respect to the length of their planning horizons:

- **Long-term planning**, or strategic planning, focuses on long-term decisions that define the structure and overall direction of the company and its supply chain. These decisions typically involve aspects such as capacity definition, facility location, and network design, and have impacts that extend over several years.
- **Medium-term planning**, or tactical planning, operates within the framework established by strategic decisions and is concerned with the coordination of regular operations. It includes

decisions related to resource allocation, production volumes, and material flows, usually within a planning horizon of several months up to two years, allowing the incorporation of demand variability and seasonal effects.

- **Short-term planning**, or operational planning, deals with short-term decisions and translates higher-level plans into detailed and executable actions. This level requires a high degree of accuracy, as it involves scheduling and sequencing activities for immediate execution, typically over a horizon ranging from a few days to a few months, and has a direct impact on service level and overall supply chain performance.

To further categorize planning tasks, they can also be divided into four functional areas, according to the nature of the task and its position in the supply chain: procurement, production, distribution, and sales. Procurement provides the required resources for production, which are then transformed into finished products and subsequently distributed to reach final customers. All these phases include short-term, medium-term, and long-term decisions (Steinrücke and Jahr, 2012).

To ensure a smooth flow of goods and information throughout the supply chain, several stages must be considered. The process typically begins with procurement, where raw materials and necessary resources are sourced and acquired. These inputs are then transferred to the production stage, where they undergo transformation into finished products. Finally, products move to the sales stage, where they are delivered and made available to customers, including activities such as marketing, pricing, and demand forecasting. From the sales perspective, accurate demand forecasting is crucial to determine production requirements. The distribution phase must also be properly defined, as it determines delivery lead times. This information serves as input for production planning, as it indicates when products are required, and, consequently, when production must be completed. Production planning then determines when to initiate the manufacturing process for each item, triggering raw material requirements and requiring coordination with procurement (Simchi-Levi et al., 2008). Conversely, the availability of raw materials constrains production activities.

Because of these interdependencies, the flow of information across the supply chain is critical and inherently bidirectional, in contrast to the flow of goods, which moves from procurement to sales, as shown in Figure 2.1 (with the exception of returns). Each stage is dependent on the others, and inefficiencies in any part of the chain can propagate and negatively affect overall performance (Sanei Bajgiran et al., 2016).

Production planning is a complex task, as the interdependencies described above create trade-offs across decisions at different stages of the supply chain. It involves managing scarce resources under multiple constraints with the primary objective of meeting customer demand efficiently and effectively (Pochet and Wolsey, 2006).

Thomas and McClain (1993) classified the key decisions in designing a production planning process into six main points. The time unit involves a trade-off: smaller units (e.g., shift-level) increase detail but also computational complexity, while larger units (e.g., monthly) reduce complexity at the expense of detail. The time horizon spans strategic, tactical, and operational levels. Regarding

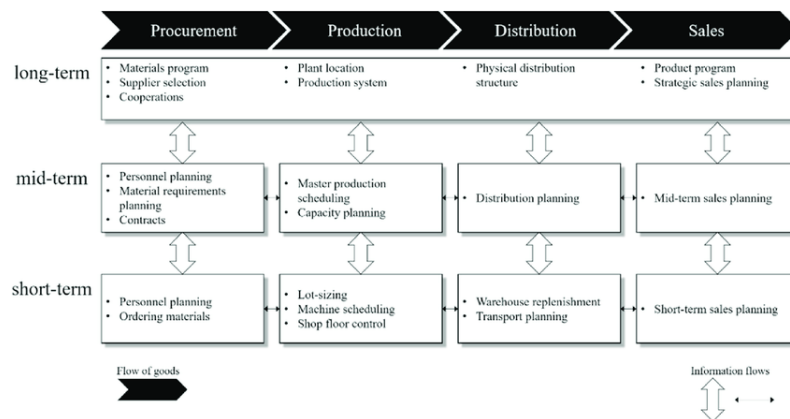


Figure 2.1: The Supply Chain Planning Matrix (Stadtler, 2005)

the level of aggregation of products, similar products can be grouped into families to smooth demand variability and reduce complexity, whereas disaggregated planning maximizes detail at a higher computational cost. Similarly, the level of aggregation of production resources consists of grouping resources by similar characteristics or functionalities to simplify capacity constraints. The frequency of replanning is typically handled through a rolling horizon framework, in which plans are periodically updated as new information becomes available. Lastly, the number and structure of production plans reflects that companies often maintain multiple interconnected plans with different levels of detail and horizons, such as aggregate tactical targets that are translated into detailed operational plans. These design decisions are fundamental when formulating mathematical optimisation models for production planning, as they directly influence model structure, level of detail, and computational tractability.

2.2 Tactical Production Planning

Tactical production planning plays an intermediary role within production planning and control systems, acting as a bridge between long-term strategic decisions and short-term operational activities. Typically addressed over a medium-term horizon, this level of planning is concerned with coordinating available resources, such as labour, production capacity, and inventory, with expected demand patterns. Its objective is to support an efficient response to market requirements while ensuring the effective use of resources across the supply chain.

As tactical planning lies between strategic and operational planning horizons, it is influenced by strategic decisions and, in turn, guides operational planning. Within this context, several key elements defined at the strategic level, including decisions related to product portfolio, network configuration, and available capacity, play a crucial role in shaping tactical production plans (Lindahl et al., 2023). Additionally, tactical planning incorporates other essential elements, including demand forecasting, resource allocation, and production scheduling, which support the development of feasible and consistent production plans.

With these inputs, tactical planning aims to determine the appropriate inventory levels for all products, and establish production targets for different facilities as key outputs (Stadtler, 2005). The main goal is to optimise resource utilization while ensuring responsiveness to demand variability. In doing so, it enables organizations to operationalize their strategic objectives and maintain alignment with evolving market conditions.

Furthermore, tactical production planning plays a critical role in supporting decision-making at the operational level by prioritizing customer orders based on criteria such as profitability, urgency, and resource availability, while also identifying and addressing bottlenecks and inefficiencies that may hinder production flow. This contributes to the reduction of lead times and the improvement of overall system performance, ultimately leading to higher customer satisfaction, which is a key determinant of organizational success.

In addition, tactical planning is concerned with minimizing production costs while maintaining product quality and ensuring that customer requirements are consistently met. These objectives reinforce its role in balancing efficiency and service levels within production systems.

Overall, tactical production planning enables the transformation of a company's strategic vision into actionable plans. Its effective implementation ensures coordinated production processes and provides a structured basis for decision-making across planning levels, contributing to improved performance and customer satisfaction.

2.2.1 Master Production Scheduling

Jacobs et al. (2011) describe the master production schedule as a statement of planned future output, representing the anticipated build schedule for the company. Master Production Scheduling (MPS) is the process of specifying how products will be supplied to meet demand. In that sense, MPS does not generate a forecast, but rather relies on demand forecasts as an input. Therefore, a master production schedule, while accounting for capacity limitations, production costs, and other relevant constraints, acts as a bridge between the market and manufacturing, connecting forecasted demand to production activities within the plant; also, it links higher-level production planning decisions with the detailed scheduling of components and raw materials (Serrano et al., 2021).

Master Production Scheduling is considered, by Soares and Vieira (2009), the most important activity in production planning and control, highlighting its critical role in guiding companies toward effective operational performance. As strategic decisions aim for long-term success of a company, tactical planning is influenced by those decisions and prepares plans that will feed more detailed operational-level plans. MPS links these planning levels, as presented in Figure 2.1, aiming to organize company's production resources, defined in the strategic level, to meet desired demand (Stadtler, 2005). Therefore, this plan affects several departments in a company. A master production schedule directly impacts the production process, affecting the production department, and is defined in order to fulfil demand, thus influencing the sales function. It supports the evaluation of trade-offs between sales and production to achieve efficient resource utilization and customer satisfaction (Jacobs et al., 2011).

Master Production Scheduling plays a critical role in improving overall operational performance, as it enables companies to define coordinated and feasible production plans. Well-structured MPS processes contribute to faster response times and economic benefits, supported by improved product quality, more efficient resource utilization, reduced lead times, and lower inventory levels (Carravilla and de Sousa, 1995). For this reason, the integration of different planning levels is essential to ensure consistency and feasibility across plans, leading companies to continuously invest in more detailed and robust MPS processes.

MPS is inherently a challenging task, as it requires the identification of production plans that are both feasible and efficient, while considering multiple constraints and trade-offs. The resulting production plan has a direct impact on key performance indicators, including production costs, inventory holding costs, and customer service levels (Comelli et al., 2008). In practice, MPS defines the production quantities of final products over time, based on both confirmed customer orders and forecasted demand, serving as a key input for downstream planning activities.

To perform this role, MPS relies on data from multiple sources across the organization, including demand forecasts, available capacity, production efficiency, inventory levels, and operational constraints. Based on these inputs, it provides key outputs such as capacity utilization plans and inventory projections over time, supporting decision-making at both tactical and operational levels. Moreover, MPS establishes a critical link between different functional areas, including sales, marketing, distribution, procurement, and production, ensuring alignment across the organization (Das et al., 2000).

The appropriate MPS strategy depends on the level of demand variability and product customization (Wortmann, 1983). Jacobs et al. (2011) identify four main production strategies, each reflecting a different position along the spectrum between full inventory anticipation and fully demand-driven production.

In a Make-to-Stock (MTS) environment, production is planned in advance based on demand forecasts, and finished goods are held in inventory from which customer orders are directly fulfilled. Service levels therefore depend on the availability of stock relative to demand, and the central planning challenge is managing the trade-off between inventory holding costs and the risk of stockouts. This strategy is best suited to products with stable, predictable demand patterns, although even then, forecast uncertainty means that safety stock buffers remain necessary.

Assemble-to-Order (ATO) occupies an intermediate position. Rather than stocking finished products, the company holds inventories of standardized components that can be combined into different end products. Production of the final item is only triggered upon receipt of a customer order, at which point the available components are assembled according to the customer's specification. The MPS in this context therefore focuses on component-level planning, and configuration management — determining which components can be combined to form which products — plays a central role.

In a Make-to-Order (MTO) environment, neither finished products nor component inventories are maintained. Although products are generally predefined, with some degree of customization, the entire production process is initiated only after a confirmed customer order is received. The

MPS in this setting is concerned with defining feasible production and delivery schedules for each individual order, making capacity management and order promising particularly critical.

Finally, Engineer-to-Order (ETO) represents the most customized approach. Once an order is placed, the product design process begins in close collaboration with the customer, and production only starts after design specifications have been fully defined and agreed. There is no pre-existing stock at any level, and lead times are typically the longest among the four strategies due to the engineering phase that precedes manufacturing.

These strategies are not mutually exclusive in practice. Many industrial companies operate under a hybrid approach, applying different strategies to different product families depending on demand predictability, customization requirements, and inventory cost trade-offs.

In summary, Master Production Scheduling plays a central role in manufacturing performance by enabling alignment across planning levels and supporting the definition of consistent and feasible production plans. By integrating information on demand, capacity, and operational constraints, MPS contributes to meeting customer requirements while maintaining operational efficiency (Soares and Vieira, 2009).

2.2.2 Capacity Planning

As stated by Jacobs et al. (2011), production planning is divided into materials planning and capacities planning. To develop a feasible and adequate production plan, it is necessary to ensure that a capacity plan is also developed and analysed. There are three horizon lengths characterizing capacity planning: short-term, medium-term and long-term, corresponding to operational, tactical and strategic planning. Chen et al. (2009) define capacity planning as the process of identifying the resources required by an organization to fulfil a given demand over a chosen planning horizon. Long-term decisions concern facility locations, plant capacities, supplier networks, and production technology, defining the structural capabilities of the system. Short-term capacity planning ensures that resource availability is consistent with the requirements of the production plan being executed at the operational level. Medium-term, tactical capacity planning sits between these two levels and addresses decisions on workforce size, raw materials, and inventory policies at the product-group or department level.

Olhager (2013) places capacity planning within the broader evolution of operations planning and control, organized around the four-level structure of S&OP, master production scheduling, materials planning, and shop floor control. Within this structure, two pure strategies are traditionally associated with the medium-term level: a chase strategy, in which capacity is continuously adjusted to follow demand, and a level strategy, in which capacity remains stable and demand variations are absorbed through inventory, backlogs, overtime, or subcontracting (Olhager, 2013; Chen et al., 2009). In practice, intermediate approaches combining both strategies are often used, since rigid adherence to either tends to be either operationally disruptive or financially inefficient.

Capacity decisions are particularly challenging because they are characterized by indivisibility, irreversibility, and non-convex costs (Van Mieghem, 2003). These features mean that capacity cannot be freely reversed once committed, often involves lumpy investments, and exhibits fixed

costs or economies of scale that complicate optimisation. In addition, capacity decisions are typically made under significant demand uncertainty, since capacity is built ahead of demand realization and cannot easily be adjusted in the short term (Smirnov et al., 2021).

To address these difficulties, a substantial part of the literature has been devoted to formalizing capacity decisions through mathematical models. Díaz-Madroñero et al. (2014), in a review of more than 250 discrete-time models for tactical production planning, observed that capacity constraints, alongside setups, backlogs, and overtime, are among the most common extensions incorporated into these formulations, with linear and mixed-integer linear programming being the dominant modelling approaches. More recent reviews, such as the one by Guzman et al. (2022), confirm this trend and highlight a growing emphasis on the integration of capacity, planning, and scheduling decisions within unified frameworks.

In practice, however, capacity and production decisions are often addressed sequentially, with capacity plans built first and production schedules adjusted to fit within them. This sequential approach frequently leads to infeasible or uneconomical schedules, motivating the development of integrated models that jointly determine capacity and production decisions (Yao et al., 2022). This limitation is particularly visible when conventional material planning approaches are used, since traditional Material Requirements Planning (MRP) systems schedule production orders under the assumption of infinite capacity and rely on fixed, predefined lead times that ignore the actual workload of the system (Jodlbauer and Strasser, 2019), as discussed in the next section.

2.3 Material Requirements Planning

MRP is built around a single calculation that answers, for any point in the planning horizon, what a company currently holds in stock, what it still has to produce or purchase, and when each of those actions must occur (Ptak and Smith, 2011). It performs this detailed material planning across the levels that connect raw materials and components to finished products, ensuring that each part is available precisely when the production schedule for the end items demands it. To do so, MRP relies on three inputs: the Master Production Schedule, the Bill of Materials (BOM), and the inventory status, as shown in Figure 2.2.

The BOM specifies, for each parent item, the components consumed directly in its production (Jacobs et al., 2011). Through the BOM, it is possible to determine the required quantity of each raw material (RM) based on the demand of finished products, recursively going down and accounting for the quantity of each component required to produce the item immediately above it, level by level, until reaching the RM, which is known as BOM explosion. Coil demand is therefore not independent demand to be forecast, but dependent demand, derived entirely from the finished-product plan: the foundational distinction on which MRP rests (Orlicky, 1975).

In order to determine the requirements for each item, MRP organizes the relevant information into a time-phased record composed of four elements: gross requirements, corresponding to the expected demand or consumption of the item in each period; scheduled receipts, which are replenishment orders already placed and scheduled to arrive at the beginning of each period; the

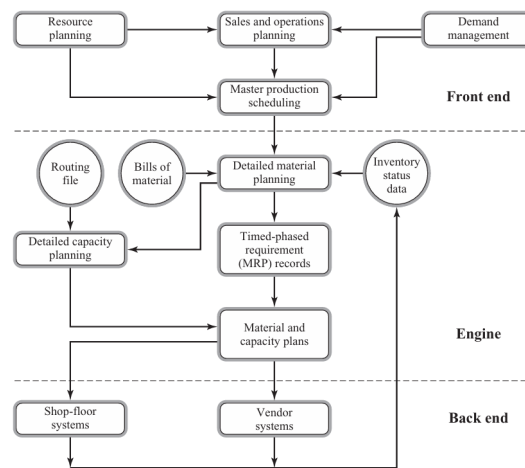


Figure 2.2: Manufacturing Planning and Control System (Jacobs et al., 2011)

projected available balance, that is, the projected inventory level of the item at the end of each period; and planned order releases, the replenishment orders that the system suggests releasing at the beginning of each period.

Each period is referred to as a time bucket, and the total number of periods constitutes the planning horizon. The gross requirements are time-phased, meaning they are expressed on a period-by-period basis, and the availability of the item determines whether a gross requirement can be satisfied in the period in which it occurs. An item is available if it is held in inventory in the required quantity, or if a scheduled receipt will arrive in time to cover the requirement. When neither condition is met, the shortfall between the gross requirement and the available inventory defines the net requirement; that is, the quantity that must be made available before the period in which it is needed (Jacobs et al., 2011).

While the net requirement specifies how much of an item is needed in a given period, it does not indicate when the corresponding order must be placed. Since every replenishment takes time to be produced or delivered, MRP applies lead-time offsetting, back-scheduling the order from the period in which the item is needed by an amount equal to its lead time, so that it arrives just in time to cover the requirement (Orlicky, 1975). The quantity to be ordered, in turn, is set by a lot-sizing rule: the simplest, lot-for-lot, orders exactly the net requirement in each period, while alternatives such as fixed order quantity or period order quantity consolidate requirements into larger, less frequent batches to reduce ordering and setup costs, at the expense of higher inventory (Jacobs et al., 2011). More elaborate approaches address this trade-off explicitly: the Wagner-Whitin algorithm computes the cost-optimal sequence of lot sizes for deterministic, time-varying demand (Wagner and Whitin, 1958), while heuristics such as the Silver-Meal method approximate this solution at a lower computational cost (Silver and Meal, 1973). The combined result of lead-time offsetting and lot-sizing is the planned order release, an order of a defined quantity scheduled for a defined period, which triggers a purchase order for bought-in materials such as raw materials.

The procedure described above rests on a set of assumptions that rarely hold in practice. MRP

treats both demand and lead times as deterministic: gross requirements are assumed to be known, and each item is offset by a single, fixed lead time (Orlicky, 1975). In reality, demand is subject to forecast error and lead times vary, and the literature has long documented how this degrades MRP performance. Mula et al. (2006) review the body of work on production planning under uncertainty and identify demand and lead-time variability as the dominant sources of instability in requirements planning. Enns (2002) shows that forecast bias and demand uncertainty propagate through the netting and offsetting logic, distorting order quantities and timing and undermining the reliability of the resulting plans. Dolgui and Prodhon (2007), in a survey specifically focused on supply planning in MRP environments, argue that lead-time uncertainty is particularly damaging and was historically underexplored. This difficulty is compounded in multi-component assembly systems where the inter-dependence of component inventories means that a delay in the supply of any single component can stall production of the parent item. A further well-known weakness is system nervousness, whereby minor revisions to the master schedule trigger disproportionate and frequent changes to planned orders, eroding the plan's stability (Blackburn et al., 1986). A second structural limitation is that classical MRP assumes infinite capacity: it explodes requirements and offsets lead times without verifying whether the implied procurement load is feasible, leaving capacity and cost considerations outside the calculation (Jacobs et al., 2011).

The model developed in this dissertation builds directly on this body of theory, while addressing the limitations identified above. Because coil demand is dependent rather than independent, it cannot be planned in isolation: following the central premise of MRP, the medium-term production model generates the coil requirements by exploding the production plan for finished products through the BOM, expressed in this work through the consumption factors that relate each component to its parent item (Orlicky, 1975; Jacobs et al., 2011). These time-phased coil requirements are then passed to a dedicated coil purchase optimisation module, which determines how each requirement should be sourced and when the corresponding order should be released. In this sense, the two modules reproduce the classical MRP information flow: the production model performs the requirements-generation logic (explosion and netting against projected coil inventory), while the purchase optimiser performs the procurement logic (lead-time offsetting and the conversion of net requirements into sized, time-phased orders).

The proposed approach departs from classical MRP precisely where the latter is weakest. Rather than offsetting requirements by a single fixed lead time and sizing orders through a predetermined heuristic rule, the purchase optimiser selects among alternative suppliers with distinct lead times and price trajectories, and decides order quantities and release moments by minimizing cost rather than applying a fixed rule. The procurement step of MRP, typically its most rigid and most criticized component, is thereby replaced by an explicit optimisation. In parallel, the limitations of deterministic demand and infinite capacity are addressed within the medium-term model itself, which incorporates safety and cycle stocks to buffer against forecast error and respects the finite production capacity that the standard MRP calculation ignores. The result is an integrated formulation that retains the requirements-generation logic of MRP while overcoming the deterministic, single-source, and capacity-unaware assumptions on which it rests.

2.4 Solving Approaches

Solving supply chain strategy problems is inherently complex. This complexity increases with the level of detail and the scale of the supply chain. Among those problems, production planning stands out, due to its high level of complexity. As organizations become more complex and specialized, the task of allocating available resources across different activities becomes increasingly difficult, particularly when aiming to achieve the best overall performance for the company (Hillier and Lieberman, 2015). To address these challenges, optimisation methods have been widely adopted, enabling the identification of feasible and optimal solutions under multiple constraints. An optimisation model is built on mathematical expressions that describe the problem's constraints and objective, which are typically represented through equalities or inequalities and an objective function (OF), respectively.

2.4.1 Different Optimisation Methods

Optimisation methods can be classified into different categories according to the focus and characteristics of the algorithm (Amiri et al., 2017). Figure 2.3 presents a possible classification of optimisation methods.

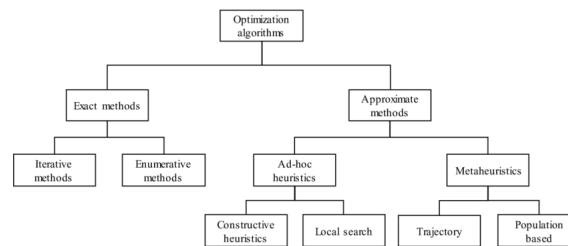


Figure 2.3: Optimisation Methods (Amiri et al., 2017)

The first division is between exact and approximate methods. As the name suggests, exact methods aim to find the optimal solution to a given problem, by maximizing or minimizing the objective function. On the other hand, approximate methods focus on finding near-optimal solutions, typically suboptimal. In general, these methods require lower computational effort and are often applied to more complex problems. When possible, exact methods are preferred due to their ability to converge to the optimal solution, whereas approximate methods do not guarantee optimality.

Inside exact optimisation methods, there are two main classes: iterative methods and enumerative methods. Iterative methods start from an initial solution and generate a sequence of improving solutions by applying mathematical rules (such as gradient- or direction-based updates), converging to the optimal solution under suitable conditions. These methods exploit the structure of the problem and are typically used for continuous optimisation. Enumerative methods systematically explore the solution space to guarantee optimality. Although they may conceptually consider all possible solutions, in practice they often use strategies such as pruning to avoid evaluating every candidate explicitly. The optimal solution is determined based on the value of the objective function.

However, the number of possible solutions grows very quickly with the size of the problem, which can lead to high computational effort. For this reason, these methods may become impractical for large-scale problems, particularly in combinatorial settings. Iterative methods are generally preferred for large-scale continuous problems, while enumerative methods are more suitable for discrete or combinatorial problems.

Inside approximate optimisation methods, two main classes can also be identified: ad-hoc heuristics and metaheuristics. Unlike exact methods, these approaches do not guarantee finding the optimal solution, but instead aim to obtain good-quality solutions within reasonable computational times. Ad-hoc heuristics are problem-specific methods that exploit the structure of a given problem to quickly construct or improve a solution. They can be divided into constructive heuristics, which build a solution step by step, and local search heuristics, which start from an initial solution and iteratively improve it by exploring nearby solutions. These methods are usually simple and efficient, but they may get trapped in local optima and are not easily generalizable to other problems. Metaheuristics, in contrast, are more general approaches designed to guide the search process across different types of problems. They are often divided into trajectory-based methods, which improve a single solution over time, and population-based methods, which work with multiple solutions simultaneously. These methods aim to balance the exploration and exploitation of the solution space, often incorporating some level of randomness to avoid local optima. Although they are more flexible and capable of producing high-quality solutions, they still do not guarantee optimality, which distinguishes them from exact methods.

2.4.2 Linear Programming

As engineering problems that aim to represent real-world systems become increasingly complex, mathematical programming has evolved accordingly, enabling the solution of more challenging optimisation problems.

Mathematical programming provides a structured framework for solving optimisation problems by determining the best possible solution subject to a set of equalities and inequalities that define the feasible solution space (Snyman and Wilke, 2018). Within mathematical programming, two main groups can be identified: linear and non-linear programming. These groups differ in the nature of the expressions used in both the objective function and the constraints. Linear programming (LP) considers exclusively linear relationships between variables, while non-linear programming admits non-linear relationships, making it generally more complex to formulate and solve (Iqbal et al., 2014).

Within both groups, models can be further divided according to the type of variables involved: continuous models allow variables to take any real value; integer models restrict variables to integer values; and mixed-integer models combine both, with at least one integer variable alongside continuous ones. The nature of the problem typically determines which formulation is most appropriate.

The origins of linear programming are attributed to Kantorovich (1939) and Leontief (1941), who independently applied linear optimisation to manufacturing scheduling and economic planning

problems. The field gained significant momentum during the Second World War through applications in logistics and resource allocation, and attracted wider academic and industrial interest in its aftermath (Hillier and Lieberman, 2015).

When integer restrictions are imposed on decision variables, the problem becomes considerably more difficult to solve. Mixed-integer linear programs (MILPs) are particularly relevant in engineering and supply chain contexts, where discrete decisions such as facility location, component selection, or binary on/off assignments are common (Scavuzzo et al., 2024). The standard solution strategy for MILPs is the branch-and-bound algorithm (Land and Doig, 1960), which recursively partitions the feasible space into subproblems, each solved as a continuous LP relaxation, and prunes branches that cannot yield an improvement over the best known integer solution. Modern solvers such as Gurobi combine branch-and-bound with cutting plane methods to tighten LP relaxations, enabling the efficient solution of large-scale MILP instances encountered in practice (Scavuzzo et al., 2024).

2.5 Overview on Theoretical Background

This chapter has framed production planning across the three time horizons that structure decision-making in industrial companies, with the medium-term, tactical level, identified as the focus of this dissertation. Tactical planning does not operate in isolation: it is shaped by strategic decisions on capacity and product portfolio, guides operational scheduling and execution, and coordinates across functional areas, from demand and master production scheduling to material requirements and procurement. Effective tactical planning therefore depends on a precise flow of information between these stages, and on the analytical tools capable of representing their interdependencies in a single, coherent decision.

In summary, MRP provides the conceptual foundation for translating finished-product demand into time-phased material requirements, and it is this logic that links the production and procurement decisions addressed in this dissertation. However, its reliance on deterministic demand and lead times, its disregard for finite capacity, and its dependence on fixed lot-sizing rules mean that the requirements it generates cannot, on their own, yield cost-effective and feasible plans in the setting under study. Overcoming these shortcomings calls for an explicit optimisation formulation rather than a sequential netting procedure, which motivates the modelling approaches reviewed in the following section.

Among such formulations, linear programming offers a structured framework for representing production planning decisions, and remains the most widely adopted approach in the field. When the problem requires discrete decisions, such as whether to set up a product, open a shift, or assign a machine, the formulation extends naturally to mixed-integer linear programming, solved through branch-and-bound and modern cutting-plane techniques. It is this class of formulations that the dissertation adopts, and on which the model developed in the following chapters is built.

Chapter 3

The Problem

This chapter aims to provide an in-depth understanding of the problem addressed in this dissertation. It begins with an overall description of the problem and the main challenges identified. The chapter then introduces the industrial and planning context, exploring the company's characteristics and operating environment. Following this, the company's production process is detailed, providing the specific context required to understand the developed solution. The proposed approach is then presented, outlining how the identified challenges will be addressed. A description of the input data structure and the data preparation steps undertaken are subsequently explored. The chapter concludes with an exploratory data analysis to support and quantify the nature of the problem described.

3.1 Problem Description

Medium-term production planning plays a pivotal role in aligning manufacturing operations with market demand and strategic business objectives. It bridges the gap between long-term strategic goals and daily operational activities, enabling companies to allocate resources efficiently, manage inventory levels, and ensure timely customer fulfilment. In practice, however, the complexity of real industrial environments often exposes significant weaknesses in the planning processes, particularly when decisions across different planning levels are made independently and without a structured analytical foundation.

At the outset of the project, a diagnostic analysis was conducted to assess the current state of planning and decision-making processes at the company. This assessment involved mapping existing planning procedures, identifying bottlenecks, and engaging with key stakeholders across production, sales, and procurement. The analysis revealed five critical pain points that collectively undermine the efficiency and effectiveness of the company's supply chain operations.

The first challenge concerns the low accuracy of sales forecasts. Commercial forecasts were found to achieve assertiveness rates of only 50% to 60% for the immediate month and as low as 20% for the following month. This high level of uncertainty propagates throughout all downstream planning decisions, making it difficult to anticipate production needs, manage inventory proactively,

or commit to delivery dates with confidence. Without a reliable demand signal, the remaining planning layers — production scheduling, raw material procurement, and inventory management — operate under significant ambiguity.

The second pain point relates to the absence of a unified inventory policy. Inventory objectives across the company were found to be fragmented, with different products and areas operating under inconsistent or ad hoc stock targets. The absence of a systematic, data-driven approach to defining safety stock and replenishment targets prevents the company from finding a coherent balance between operational efficiency and customer service, ultimately leading to suboptimal inventory levels across the factory.

The third challenge identified concerns supplier evaluation practices that rely predominantly on acquisition price, without considering a broader set of relevant factors. Costs associated with holding inventory, risks of supply delays, and supplier reliability were not systematically incorporated into sourcing decisions. As a consequence, purchasing choices that appear cost-effective in isolation may generate hidden costs downstream, such as emergency orders, production stoppages due to material shortages, or excess stock resulting from bulk purchasing driven purely by price incentives. This myopic approach to procurement limits the company's ability to manage its supply chain as an integrated system.

The fourth pain point is the pressure to reduce delivery lead times in the absence of adequate planning visibility. Customer expectations regarding delivery responsiveness have tightened, yet the internal process for committing to and approving delivery dates is slow and poorly integrated with production planning. The lack of a structured, optimisation-based tool means that decisions about accepting or scheduling orders are made reactively, based on individual judgment rather than a global assessment of capacity and inventory availability. This creates a recurring tension between the commercial drive to shorten lead times and the operational constraints that govern what is feasible — a trade-off that is currently handled empirically rather than analytically.

The fifth and final pain point concerns small production lots and the resulting reduction in operational efficiency. The current planning approach does not systematically account for the cost of setups, which are significant in the metal packaging industry given the complexity of changeovers between different products and packaging formats. As a result, production runs tend to be fragmented, with frequent setups that consume productive capacity and increase unit production costs. This inefficiency is compounded by the fact that lot-sizing decisions are not made within an optimisation framework that explicitly balances setup costs against inventory holding costs — a trade-off that is central to any effective tactical production planning approach.

Taken together, these five pain points reflect a planning environment that lacks integration, analytical rigor, and the decision-support tools necessary to coordinate demand, production, and procurement in a coherent and cost-effective manner. The development of an optimisation-based planning framework, as pursued in this dissertation, directly responds to these challenges by providing a structured, data-driven approach to medium-term production decisions.

Addressing each of these pain points in isolation, without a holistic view of the company's operations, risks worsening other areas of the business. Increasing inventory levels may resolve

stockout issues but simultaneously inflates holding costs. Similarly, sourcing from alternative local suppliers may reduce raw material lead times but typically comes at a higher purchase price. This inherent tension between competing objectives is precisely what makes an optimisation-based approach necessary — one that evaluates costs and trade-offs in a unified framework, seeking the best overall solution for the company rather than locally optimal decisions that shift inefficiencies from one area to another.

3.2 Industrial and Planning Context

The company under study is a Brazilian metal packaging manufacturer, with more than one century of history, that produces metal cans for two main industries: food and chemicals. In the food segment, the company serves clients requiring packaging for products such as vegetables, fruits, and meats, while in the chemical segment, it produces cans for varnishes, glues, adhesives, and paints. Since all packages can undergo a lithography process, from which they emerge with printed logos and designs specific to each client, the production process results in fully customized end products. As a consequence, the relationship between finished products and customers is many-to-one: a customer may have multiple products in the company's portfolio, but each finished product is exclusively associated with a single customer.

The company operates in the B2B (business-to-business) market, supplying metal packaging to other companies rather than directly to end consumers. With a customer portfolio with a size greater than 100, it is a well-established player in the Brazilian metal packaging industry. The product itself does not hold intrinsic value to the final consumer — it serves as the container for the value-adding product it holds — yet it remains essential, as without it the final product cannot be delivered to market. Operating across two main manufacturing facilities, the company has the capacity to process more than six thousand tons of steel coil per month. The present study focuses exclusively on one of these plants, the main one, hereinafter referred to as Factory 01.

Factory 01 produces not only to fulfil customers' demand, but also to supply intermediate components required by the second plant. In some cases, therefore, the production process does not conclude at Factory 01 — instead, semi-finished products are transferred to the second facility for finalisation and subsequent delivery to customers. The plant is organized into three main areas: lithography, stamping, and assembly, which will be described in further detail in Section 3.3.

Regarding production strategy, the company does not follow a single unified approach. Although all finished products are customized, production is governed by two strategies: Make-to-Stock (MTS) and Make-to-Order (MTO), with more than 85% of products being produced under the latter. MTS production can take two forms: in some cases, the customer shares a demand forecast for the upcoming period and the company produces only the quantities required to fulfil that plan; in others, no forecast is provided and the company relies on its own demand estimates to decide how much to produce. In both cases, it remains a MTS approach in the sense that customers do not place orders in advance — the decoupling point lies at the finished goods inventory, from which customers

purchase directly. This model entails a degree of loyalty between the company and its customers, since, as previously noted, each finished product can, usually, only serve a single customer.

The current planning process is partially supported by the company's Enterprise Resource Planning (ERP) system, but relies heavily on empirical knowledge and spreadsheet-based tools to fill the gaps that the system does not address. Planning decisions across different horizons — from raw material procurement to production scheduling — are made without a systematic, integrated framework, with each decision converging to a local optimum that, in the absence of a globally coordinated approach, can yield suboptimal or even severely detrimental results at the system level. The absence of an optimisation-based planning tool means that trade-offs between costs, capacity, and service levels are evaluated informally, which, as discussed, is at the root of several of the pain points identified during the diagnostic phase. The scale and complexity of the planning problem in Factory 01, in terms of product portfolio, customer base, and product structure, are characterized in detail in Section 3.7 through an exploratory analysis of the available data.

3.3 Production Process

The production process at Factory 01 begins with the receipt of steel, which can be sourced either as coils or as pre-cut sheets, depending on the supplier. When steel arrives in coil form, it is first processed through an industrial steel cutting machine, which slits and cuts the coil into flat sheets of standardized dimensions. The resulting product at this stage is referred to as a Type 01 sheet.

Regardless of the input format, all sheets subsequently undergo a varnishing process, in which a protective coating is applied to the steel surface, yielding what are internally classified as Type 02 varnished sheets. This step is particularly critical for food-grade packaging, where direct contact between the metal and the product contents demands strict quality and safety standards.

Following varnishing, the production flow diverges according to the intended use of each sheet, as illustrated in Figure 3.1. Sheets destined for can bodies proceed to the lithography process, where printed designs and logos are applied to the steel surface, producing Type 03 printed sheets. This step reflects the fully customized nature of the finished products. The lithography area is equipped with different types of printing machines: quadricolor printers, capable of applying four colors in a single pass, and two-color printers, which require multiple passes to achieve the same result, implying additional processing time and operational complexity. Although rare, some body sheets may bypass the lithography step when no printed design is required.

Sheets intended for the production of tops, bottoms, and other accessory components, on the other hand, do not typically undergo lithography, as these components are generally universal — that is, they are not customer-specific and can be incorporated into multiple different finished products. These components are produced through a stamping process, in which flat steel pieces are pressed into their final three-dimensional shape, and are classified as Type 05 intermediates.

After lithography, each Type 03 printed sheet contains multiple body outlines that must be separated into individual pieces through a secondary cutting operation, yielding Type 04 cut bodies. These individual pieces are then formed into cylindrical can bodies. It is worth noting that the

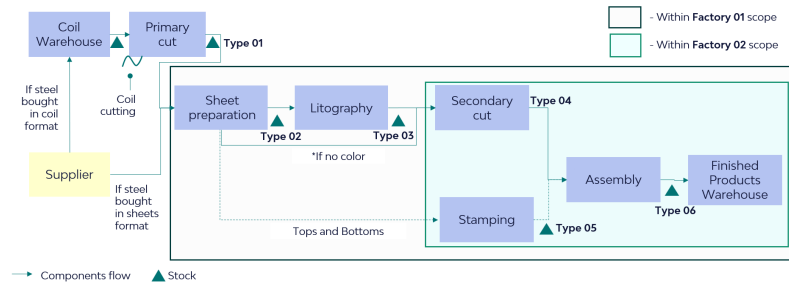


Figure 3.1: Production process flowchart

production process involves several intermediate storage points between stages, as highlighted in Figure 3.1, which introduce variability and uncertainty into the planning process — particularly regarding the timing and quantities of materials available at each subsequent stage.

In the final assembly stage, Type 04 can bodies are brought together with Type 05 tops and bottoms to produce Type 06 finished products, which are then transferred to the finished goods warehouse. In some cases, however, the production process does not conclude at Factory 01. Certain intermediate products — Type 02, Type 03 and Type 05 — are transferred to the second manufacturing facility, where they complete the remaining production steps before being delivered to the customer.

In terms of factory layout, Factory 01 is organized into three main production areas. The lithography area encompasses sheet preparation, varnishing, and the printing process itself. The stamping area is dedicated to the production of tops, bottoms, and other accessory components. Finally, the assembly area brings together the previously produced components to build the finished can, and also includes the secondary cutting operations that precede body forming.

3.4 Proposed Approach

To tackle the previously described problems, the project was structured around the development of four different yet interconnected modules, each aiming to address the five identified pain points, explored in Section 3.1, in a distinct way. As presented in Figure 1.1, the first module consists of a predictive model based on Time Series Analysis (TSA) and Machine Learning (ML) methods. Time Series Analysis comprises a set of statistical techniques for modelling data points indexed in chronological order, with the goal of identifying underlying patterns — such as trend, seasonality, and cyclical behaviour — in order to forecast future values from historical observations (Hyndman and Athanasopoulos, 2021). Machine Learning, in turn, refers to a class of computational methods that learn relationships directly from data without being explicitly programmed for the task, building models that generalise to previously unseen observations (Hastie et al., 2009; Bishop, 2006). This module directly addresses the first identified pain point: the low accuracy of sales forecasts.

Beyond its direct impact on forecast accuracy, the demand planning module plays a foundational role in the project as a whole. It serves as the baseline upon which all subsequent modules depend,

making it not only a direct response to one of the five identified pain points, but also the enabler of a structured approach to tackling the remaining ones.

The second module, and the core focus of this dissertation, is the medium-term production planning model. Fed by the outputs of the demand planning module, it determines what to produce, in what quantities, on which resources, and when — at a weekly granularity over a planning horizon of 32 weeks. The outputs of this module serve as inputs to the two remaining modules: short-term production planning and coil purchase optimisation. Beyond its primary planning function, this module also contributes to addressing several of the identified pain points. By explicitly evaluating the trade-off between inventory holding costs and setup costs, it provides a structured foundation for a unified inventory policy — one that can leverage the flexibility offered by universal intermediate components, such as tops and bottoms, which can serve multiple finished products. The pressure to reduce customer lead times is also partially addressed at this stage, as the model provides visibility over future production capacity and inventory availability. Furthermore, by optimising lot sizes within a cost-minimisation framework, it counteracts the inefficiency associated with excessively small production runs.

The third module, short-term production planning, is fed by the outputs of the medium-term model and operates at shift-level granularity. Its purpose is to translate the macro-level plan into an actionable operational schedule, guiding daily production decisions at the plant level.

Finally, the fourth module addresses the pain point related to supplier evaluation and coil purchase decisions. An optimisation model was developed to improve procurement decisions, incorporating a broader set of cost factors — including holding costs and supply risk — beyond the acquisition price alone. This module is also fed by the outputs of the medium-term production planning model, ensuring that procurement decisions are aligned with the anticipated production needs.

3.5 Input Data Structure

The optimisation model draws its inputs from two main sources: the company's data warehouse, which stores both operational and transactional data extracted from the ERP system, and a dedicated web application (WebApp) developed as part of the project, which allows the planning team to configure parameters that are not available in the data warehouse or that require periodic adjustment.

Data in a corporate data warehouse is typically organized into two types of tables (Kimball and Ross, 2002). Dimension tables contain descriptive attributes of the business entities — such as products, machines, and routings — and are generally wide in terms of columns but shallow in terms of rows. Fact tables, on the other hand, record measurements and transactional events — such as production orders, stock levels, and demand forecasts — and tend to grow over time as new records are appended. Together, these two table types form the foundation upon which the model's input dataframes are built.

The central dimension table for this model is the product master, in which each product is identified by a unique product code. A key characteristic of the production process is that a product

receives a new code each time it undergoes a transformation step, meaning that the same physical item is represented by different codes at different stages of the Bill of Materials (BOM). To facilitate identification, product codes follow a prefix convention that encodes the product type, as described in Table 3.1.

Table 3.1: Product type classification by code prefix

Prefix	Product Type
01	Steel sheets (output of coil slitting and cutting)
02	Varnished sheets
03	Lithographed sheets
04	Cut lithographed bodies
05	Stamped components (tops, bottoms, and accessories)
06	Finished metal cans
07	Inks and varnishes (secondary raw material inputs)

The routings table defines the production sequence for each product, associating each transformation step with the machines capable of performing it. For each product-machine combination within a routing, this table specifies the required workforce, the unit production time (inverse of cadence), and the setup time — both theoretical time values, not used in the model.

The machine master table describes the available production resources at Factory 01, specifying for each machine its description and the production area to which it belongs — lithography and varnishing, stamping, or assembly.

The BOM is defined in a dedicated table that specifies, for each parent product, its components and the quantity of each component required to produce one unit of the parent. This table also captures the structure of optional component groups: for certain products, a group of alternative components exists from which any one can be selected to fulfil the same role in the assembly. The model respects this flexibility by enforcing the total consumption across all components of a group, rather than fixing a specific component choice.

From the fact tables, the most relevant inputs are the following. The initial stock fact table records the current inventory level of each product across the warehouses of Factory 01, providing the starting inventory for the first week of the planning horizon. The historical production orders fact table records all past production events, including quantities produced, production times, and scrap quantities; as described in Section 3.6, this table is the source from which setup times, actual cadences, and per-unit scrap rates were initially estimated. Finally, the demand planning output fact table — produced by the first module of the project — contains, for each product and week, the confirmed order quantities, the predicted demand, and the safety stock targets that serve as the primary demand inputs to the optimisation model.

Parameters that are not stored in the data warehouse, or that require periodic adjustment by the planning team, are configured through the WebApp. These include: warehouse storage capacities;

penalty costs for unmet confirmed orders, unsatisfied predicted demand, cycle stock violations, and safety stock violations; setup costs and the Weighted Average Cost of Capital (WACC), used to compute inventory holding costs; the objective function weights for service level, purchase, stock, setup, and outsourcing cost components; the shift calendar for each production area, specifying the number of available working days per shift per week; workforce availability per crew group and shift; machine unavailabilities due to planned maintenance or other operational factors; setup times and production cadences; outsourced production machines, along with their capacities and costs; a steel coil procurement calendar, which specifies the possible purchase dates for the primary raw material from each supplier, as well as the corresponding delivery dates and total costs, and procurement lead times, defined globally with the option to specify by customer or raw material.

3.6 Data Preparation

The quality of any optimisation model's outputs is fundamentally constrained by the quality of its inputs. A model can be rigorously formulated and technically sound, yet yield unreliable results if fed with flawed data — a principle commonly known as "garbage in, garbage out". Consequently, data preparation represents one of the most critical stages in the development of a production planning optimisation model.

The significance of this stage is well recognized in the analytical literature. The Cross-Industry Standard Process for Data Mining (CRISP-DM), a widely adopted methodology for structuring analytical projects whose six phases are depicted in Figure 3.2, consistently identifies data preparation as the most time-consuming phase of any data-driven project. In line with this, Nisbet et al. (2018) states that between 60% and 90% of a project's duration is typically devoted to data preparation activities.

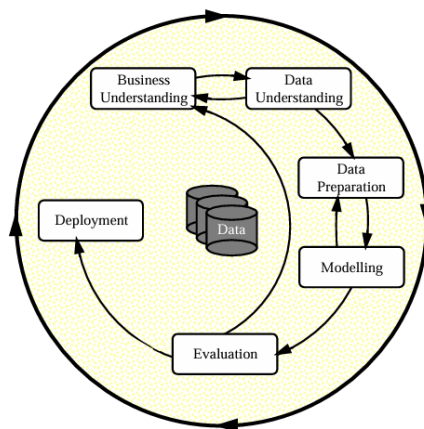


Figure 3.2: CRISP-DM Process Model (Wirth and Hipp, 2000)

In practice, real-world data is rarely clean or complete. Common issues include missing values, outliers, and inconsistencies across data sources. Each of these must be carefully identified and addressed before the data can serve as model input, as any undetected problem at this stage will

inevitably propagate into the model's outputs and compromise the reliability of the generated production plans.

In addition to cleaning and correctly preparing data, it is also important to select only the data that is strictly necessary. Including unnecessary data increases the model's size and computational burden, which can degrade solver performance and compromise the quality of results.

From the demand planning module, a dataframe containing the codes of the finished products for which demand is planned is delivered. Associated with each product is the week to which the demand applies, disaggregated into already confirmed orders and overall demand. Safety stock targets, also computed in the demand planning module, are likewise provided as inputs to the medium-term planning model for finished products.

As previously described, this project focuses on Factory 01. However, some finished products require a final processing stage carried out in Factory 02. Consequently, an additional data treatment step is required for the outputs of the demand forecasting module before they can serve as model inputs.

The first step of this treatment consists of a recursive BOM explosion, starting from the finished products and traversing the component tree downwards, in order to identify all products that are connected to those in scope. This process determines the complete set of products to be represented in the model.

Since the model is scoped exclusively to Factory 01, the demand signals associated with finished products completed in Factory 02 — namely confirmed orders, predicted demand, and safety stock — cannot be directly used. Instead, these signals must be transferred to the last intermediate product in the production chain that is manufactured in Factory 01, which feeds into the finishing operations performed in Factory 02. Furthermore, the lead time between the two factories must be accounted for: each demand signal is shifted back in time by the corresponding inter-factory lead time, ensuring that the intermediate product is available in Factory 01 with enough anticipation to guarantee on-time delivery of the final order to the customer.

Through this treatment, the model is supplied exclusively with products whose production is planned within Factory 01: finished products manufactured entirely in Factory 01, the last intermediate product produced in Factory 01 for finished products that undergo further processing in Factory 02, and all components associated with those products. This ensures that the model is not populated with products outside its planning scope.

Since all finished products are fully customized, excess stock production carries a significant risk of obsolescence. For MTO products, no inventory is held between orders, as production is only initiated upon receipt of a confirmed customer order. For MTS products, production is carried out in anticipation of demand, and the customer order decoupling point is the finished goods stock. In this case, each MTS product is associated with a single dedicated client, meaning that production requires prior authorization from that client.

As a consequence, stock levels for MTS products must not exceed the client-authorized quantity, as producing beyond this threshold creates the risk of holding inventory that the client may ultimately

not purchase, resulting in obsolete stock. This constraint is therefore incorporated into the data preparation stage as an upper bound on the stock level for each MTS product.

The derivation of the stock cap parameter $Cap_{p,w}$ requires three preprocessing adjustments. The cap is defined as the quantity the client is expected to purchase in the future: if a client forecasts 1000 units for the upcoming period and has already purchased 500, the maximum stock that can be held is 500 units. First, the cap for each week is computed as the maximum between the initial stock and the remaining authorised quantity after deducting cumulative demand up to that week — the initial stock floor ensures that the cap never falls below the stock already held, since existing inventory cannot be undone. Second, the minimum lot size is considered: since production must occur in quantities at or above the minimum lot size, a production run may push the closing stock above the raw cap. To accommodate this structurally unavoidable overshoot, the cap is relaxed upward by the minimum lot size wherever the combination of current stock, demand, and minimum lot size would otherwise make the constraint infeasible. For example, if the cap is 100 units, current stock is 50, weekly demand is 70, and the minimum lot size is 200, the model must produce at least 200 units, yielding a closing stock of 180 — 80 units above the raw cap. The adjusted cap of 180 accommodates this unavoidable overshoot while still preventing discretionary overproduction beyond it.

For MTS products, the cycle stock is calculated based on the following week's demand, ensuring that, at the end of each week, the company holds sufficient inventory to fulfil the demand of the subsequent period. Both cycle stock and safety stock of finished products can be split between intermediate lithographed sheets (product Type 03) and fully finished products (product Type 06), reflecting the two stages at which inventory can be held before delivery. This distinction is relevant because it is precisely at the lithography stage that the product becomes fully customer-specific — prior to printing, sheets are universal and can serve multiple finished products, but once printed, they are irrevocably associated with a single customer.

For intermediate products, the cycle stock is also derived from the demand of the following week, but adjusted for each level of the BOM. Specifically, at each step down the BOM hierarchy, the cycle stock is shifted one week earlier, to ensure that, when only one production stage can be completed per week, the required inputs are available at the right time. These calculations rely on the BOM consumption factors: if a finished product A requires five units of component B one step down in the BOM, and component B only serves product A, then the cycle stock of component B is five times the next week's demand of product A.

Since the model only represents finished products for which confirmed orders, demand forecasts, and safety stock targets are defined — along with all their related components — stocked products with no connection to any planned finished product are excluded from the model to avoid unnecessarily enlarging its scope. However, these excluded products physically occupy warehouse space and must therefore be accounted for in the capacity constraints. To handle this, a preprocessing step identifies all such products and deducts their current stock quantities from the available capacity of the respective warehouses before the model is solved.

Overall Equipment Effectiveness (OEE) is a composite performance indicator widely used in

manufacturing to measure how effectively production equipment is being utilized. It incorporates three dimensions: Availability, which captures the proportion of planned production time during which the equipment is actually operational; Performance, which measures how closely the actual production rate approaches the theoretical maximum rate; and Quality, which reflects the proportion of output that meets specification without requiring rework or resulting in scrap (Mitsel et al., 2021). Incorporating OEE into a production planning optimisation model is essential to ensure that the production times and capacities fed into the model reflect realistic operating conditions rather than theoretical maximums.

Accurately estimating setup times and production rates per product-machine combination is a critical data preparation step, as these parameters directly govern the capacity constraints of the optimisation model and operationalise the Availability and Performance dimensions of OEE. Although the company's ERP system stores target values for setup times and unit production times, these represent theoretical expectations rather than actual observed performance. Since real operating conditions — such as operator variability, machine wear, and product-specific complexity — systematically cause deviations from those targets, using ERP values directly would lead to overoptimistic capacity estimates and infeasible production plans. For this reason, an estimation approach based on historical production order data was pursued, as described in the following paragraphs; however, as the results obtained were not satisfactory, both parameters were ultimately defined as user-configurable inputs in the WebApp. Regarding the Quality dimension of OEE, the scrap rate is represented in the model through a per-unit component loss factor: for each parent product-component pair, the average quantity of component scrapped per unit of parent product produced is computed as a weighted average across all historical production orders, and added to the nominal BOM consumption factor, ensuring that the material requirements fed into the model account for expected production losses.

The estimation approach is based on a linear programming model fitted independently for each machine, following the structure proposed in Equation 3.1. For each recorded production order i , the observed total production time is decomposed into a setup component and a run component:

$$\hat{t}_i = z_i \cdot s_p + q_i \cdot u_p \quad (3.1)$$

where $z_i \in \{0, 1\}$ is a setup flag that equals 1 when the product changes between consecutive orders on the same machine, s_p is the setup time for product p , q_i is the quantity produced in order i , and u_p is the unit production time for product p on that machine. The setup flag is automatically derived from the production history by sorting orders chronologically per machine and detecting product transitions. The model minimises the Mean Absolute Error (MAE) between observed and estimated production times across all orders for a given machine, with s_p and u_p as decision variables, both constrained to be non-negative. When a theoretical standard time exists for a product, it is enforced as a lower bound on u_p , preventing physically implausible estimates. The production rate (cadence) for each product-machine combination is then obtained as $1/u_p$.

Before fitting, two data quality treatments are applied. First, observations with a recorded unit

rate faster than the theoretical standard time are removed, as these likely reflect data entry errors or incomplete records. Second, an interquartile range (IQR) filter is applied to non-setup observations to remove upper outliers in the unit rate distribution. Additionally, for products with fewer than three observed setups on a given machine — insufficient to identify a product-specific setup time — a shared setup time variable is used across all such sparse products on that machine, preventing overfitting to noise.

The results obtained were not satisfactory, as the production order records stored in the ERP system proved unreliable for this purpose: recorded order durations were inconsistent, with setup time included in some instances and omitted in others. As a consequence, both setup times and production rates were defined as user-configurable inputs in the WebApp, where the planning team can introduce and refine these values based on operational knowledge.

Finally, valid product-machine-routing combinations are filtered to include only those pertaining to Factory 01, ensuring that no capacity or routing data from other facilities enters the model.

Beyond the physical and operational parameters described above, the model also requires cost and penalization parameters that reflect the company's planning priorities. These include penalty weights for unmet confirmed orders, unsatisfied forecast demand, and safety stock violations, as well as setup costs, inventory holding costs, coil purchase costs and outsourcing costs. These parameters are loaded from a dedicated configuration table, allowing the planning team to adjust the relative importance of each objective component without modifying the model itself.

3.7 Exploratory Data Analysis

To better understand the context and complexity of the planning problem, an exploratory analysis of the available data was conducted. Data constitute a fundamental source of information, and the company generates extensive operational records associated with every production and commercial activity performed.

As mentioned above, this study focuses on Factory 01. This factory is responsible for the production of more than 700 finished products, and also supplies intermediate components required for the production of finished products manufactured exclusively at Factory 02. Factory 01 is therefore the core of the company's production operations.

3.7.1 Data Outline

The finished product portfolio of the company comprises 1,058 active SKUs, served to 124 active customers. Beyond finished products, the production process involves a large number of intermediate products across five distinct types, reflecting the multi-level nature of the BOM. In total, the system encompasses more than 4,700 active items across all product types, underscoring the scale and complexity of the planning decisions involved.

The production equipment is distributed across four main areas. The varnishing area operates 3 machines, the lithography area is equipped with 5 printing machines, the stamping area contains

16 stamping presses, and the assembly area has 6 assembly lines. The medium-term production planning model developed in this dissertation covers a horizon of 32 weeks at weekly granularity.

3.7.2 Demand Characterisation — ABC-SEIL Analysis

To better understand the structure and predictability of demand, two complementary classification techniques were applied to the historical demand data at the finished product level: an ABC analysis and a SEIL classification. Both were computed using total quantity as the classification metric, as this dimension is the most directly relevant to a production planning context, where capacity is consumed and lot-sizing decisions are made in physical units.

The ABC analysis classifies products according to their relative contribution to total demanded quantity. Following the conventional thresholds, products accounting for the first 80% of cumulative quantity are classified as A, those between 80% and 95% as B, and the remaining as C. The result, presented in Figure 3.3, reveals a pronounced Pareto effect: approximately 8% of the products (class A) concentrate 80% of the total quantity, while at the opposite end, around 70% of the products (class C) account for merely 5% of the volume. This strong concentration indicates that a relatively small subset of products drives the bulk of the production workload, an important consideration when prioritising planning effort and managing service levels.

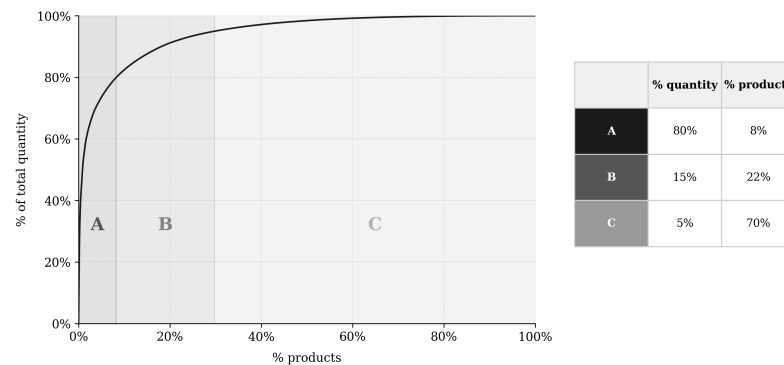


Figure 3.3: ABC product segmentation by quantity sold

To complement the ABC classification, a SEIL analysis was carried out at week granularity, categorising products according to their demand pattern as Smooth, Erratic, Intermittent, or Lumpy. This classification is based on two factors: the average demand interval (ADI), which captures intermittency, and the squared coefficient of variation (CV^2), which captures variability in demand size. Following Syntetos et al. (2005), the thresholds of 1.32 for the ADI and 0.49 for the CV^2 were adopted. Products with low intermittency and low variability are classified as Smooth, those with high variability but low intermittency as Erratic, those with low variability but high intermittency as Intermittent, and those exhibiting both high variability and high intermittency as Lumpy.

The combined results of both classifications are presented in Figure 3.4. The most striking finding is the overwhelming predominance of lumpy demand: approximately 99% of the products are classified as Lumpy, jointly representing over 96% of the total demanded quantity. Moreover,

this pattern persists even among the most commercially significant products, with class A products being almost entirely lumpy — Lumpy-A products alone account for 76.3% of the total quantity. Smooth, Erratic, and Intermittent patterns are, by contrast, negligible, both in terms of number of products and quantity.

	A	B	C	
Smooth	2.2% 0.1%	0.2% 0.2%	0.0% 0.0%	2.3% 0.3%
Erratic	1.5% 0.6%	0.1% 0.1%	0.0% 0.1%	1.6% 0.8%
Intermittent	0.0% 0.0%	0.0% 0.0%	0.0% 0.0%	0.0% 0.0%
Lumpy	76.3% 7.5%	14.8% 21.3%	5.0% 70.1%	96.1% 98.9%
	80.0% 8.2%	15.0% 21.5%	5.0% 70.2%	100.0% 100.0%

Each cell shows two values: the top (bold) is the % of quantity and the bottom is the % of products falling in that ABC-SEIL combination. The right column and bottom row give the SEIL and ABC marginal totals, respectively, and the bottom-right corner is the grand total (100%).

Figure 3.4: ABC-SEIL analysis matrix of products by quantity sold

This characterization carries direct implications for the planning problem addressed in this dissertation. The combination of a strong Pareto concentration with a demand profile that is almost entirely lumpy reflects an environment of high demand uncertainty and irregularity, which is inherently difficult to plan for using empirical or rule-based methods. This reinforces the relevance of the first pain point — the low accuracy of demand forecasts — and motivates the need for a robust, optimisation-based medium-term planning approach capable of accommodating such variability while balancing production efficiency and service levels.

3.7.3 Demand Patterns

To complement the portfolio characterisation performed through the ABC-SEIL analysis, the temporal distribution of demand was examined at an aggregate level. Figure 3.5 presents the monthly evolution of total demand throughout 2025, expressed as a percentage of the peak month. The data reveals a clear seasonal pattern, with two distinct demand peaks across the year. The primary peak occurs in May, which represents the maximum demand month, followed by a secondary peak in November, reaching approximately 81% of the May level. Conversely, demand reaches its lowest points in January (36% of peak) and in August and September (43% and 46% of peak, respectively). The trough observed in August and September is consistent with the winter period in the southern hemisphere, during which industrial activity in Brazil typically slows due to seasonal holidays and reduced consumption. The January trough, in turn, reflects the start-of-year period, which in Brazil is associated with the summer holiday season and a consequent reduction in commercial activity.

The amplitude of this seasonal variation is substantial: the ratio between the peak month and the lowest month is nearly three to one, meaning that the factory must be capable of handling

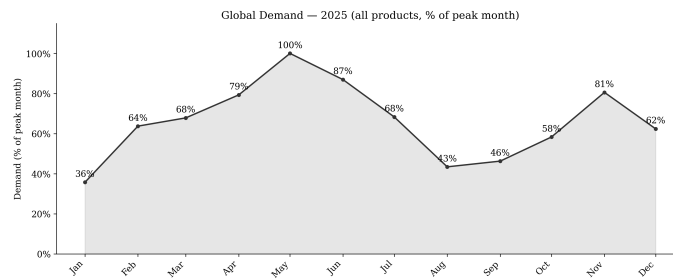


Figure 3.5: Global demand evolution — 2025 (all products, % of peak month)

production volumes that are almost three times higher in peak periods than in the trough. This level of variability has direct implications for medium-term production planning. A planning horizon that does not extend sufficiently far into the future would fail to anticipate these peaks, leading to reactive and suboptimal capacity and inventory decisions. This observation supports the choice of a 32-week planning horizon for the model developed in this dissertation, ensuring that the planning window is wide enough to capture at least one full seasonal cycle.

3.7.4 BOM Complexity — Optional Components

The BOM structure of the company introduces an additional layer of planning complexity, which was examined through an analysis of the optional component groups present in each finished product. An optional group is defined as a set of alternative intermediate components from which exactly one must be selected in order to produce a given finished product. The existence of optional groups reflects the flexibility embedded in the production system — a flexibility that, if properly exploited, can be leveraged to reduce inventory requirements and improve responsiveness to demand.

The analysis, presented in Figure 3.6, reveals that the portfolio contains 429 optional groups in total, with a median group size of 2 alternatives and a mean of 3.0. The majority of groups are small: 75% have 3 or fewer alternatives, although some reach up to 12, indicating that for certain intermediate components, a significant number of finished products can be served from the same upstream stock. This confirms the relevance of the second pain point discussed earlier — the absence of a unified inventory policy — as the existence of these shared components creates a natural opportunity for pooled inventory management that the current planning approach does not systematically exploit.

At the finished product level, the complexity is equally pronounced, as shown in Figure 3.7. Each finished product has a median of 7.5 optional groups, meaning that, typically, 7.5 independent component selection decisions must be made for each product. The distribution is wide, ranging from products with no optional groups to products with up to 19. When the number of optional groups is combined with the size of each group, the resulting configuration space — that is, the total number of possible component combinations for a given product — grows exponentially. As illustrated in the scatter plot, products with 15 or more optional groups already exhibit configuration

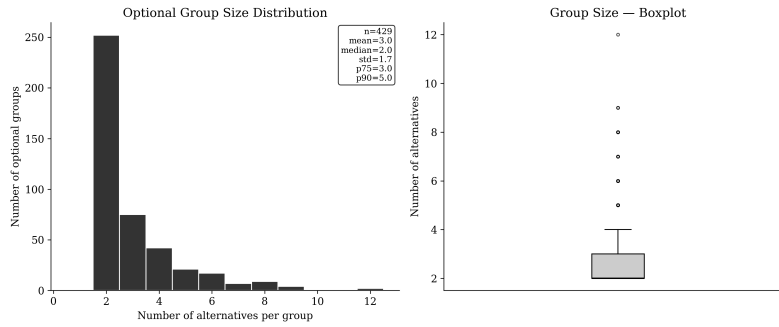


Figure 3.6: Optional group size distribution and boxplot

spaces exceeding 2^{20} , and the most complex products surpass 2^{30} possible combinations. This combinatorial explosion makes it computationally infeasible to enumerate all possible configurations manually, further reinforcing the necessity of a structured, optimisation-based planning approach.

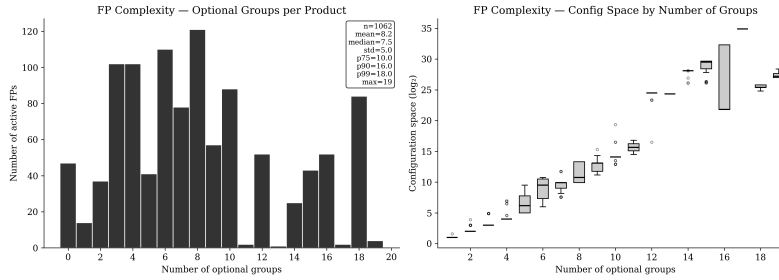


Figure 3.7: Finished product complexity: optional groups per product and configuration space

Taken together, the analyses presented in this section paint a quantitative picture of a planning environment characterised by high portfolio complexity, highly irregular and seasonal demand, and a rich BOM structure with significant shared component flexibility. These findings collectively motivate the development of a mathematical optimisation model capable of handling this complexity in a structured and systematic way, as described in the following chapters.

3.8 Problem Overview

The problem addressed in this dissertation encompasses the challenges of optimising a production process — explored in Section 3.3 — characterised by multiple sequential stages, a multi-level BOM, and groups of optional components. The quantitative dimension of this problem is characterised in Section 3.7, illustrating the scale of the portfolio, the irregularity and seasonality of demand, and the structural complexity embedded in the BOM.

However, the challenge extends beyond the production process itself. Building an optimisation model requires mapping, cleaning, and preparing all relevant data to serve as reliable inputs — a process described in Section 3.6 that is as demanding as the mathematical formulation. Together, these elements define a problem that is both practically relevant and methodologically challenging, motivating the optimisation-based approach formally developed in the following chapter.

Chapter 4

Methodology

This chapter presents the methodology developed to address the problem described in Chapter 3. It begins by establishing the model's features and assumptions, framing the scope and the simplifying choices adopted in the formulation. The chapter then introduces the mathematical formulation of the model, starting with the sets, subsets, parameters, and decision variables that define its notation. The objective function is presented next, followed by the full set of constraints, organised into three groups: inventory and material flow, capacity, and confirmed orders fulfilment. The chapter concludes with a summary of the model, connecting its formulation back to the planning challenges identified in Chapter 3.

4.1 Model Features and Assumptions

To address the planning challenges described in Chapter 3, a Mixed-Integer Linear Programming (MILP) optimisation model was developed. The model minimizes a weighted sum of costs encompassing five components: service level costs — associated with unmet confirmed orders, unsatisfied predicted demand, and violations of safety stock and cycle stock targets —, steel coil purchase costs, inventory holding costs, setup costs, and outsourcing costs. Each cost component is weighted by a configurable priority factor, allowing the planning team to adjust the relative importance of each objective component to reflect the company's operational priorities without modifying the model itself.

The model operates over a planning horizon of 32 weeks, with a weekly time bucket, covering approximately eight months of forward visibility. It is run on a weekly basis following a rolling horizon approach, meaning that each week the model is re-solved with updated inputs — refreshed demand signals, current inventory levels, and revised confirmed orders — producing an updated production plan for the following 32 weeks.

Within this horizon, the model simultaneously determines three sets of decisions. The first concerns production: for each product, routing, and week, the model decides whether to produce and in what quantity, while respecting minimum lot sizes and the available machine capacity defined by the company's shift calendar. The second concerns inventory: stock levels for all products —

finished goods, intermediate products, and raw materials — are tracked week by week across the full horizon, ensuring that material flows across Bill of Materials (BOM) levels are respected. The third concerns procurement: the model determines the quantity of steel coil to purchase in each week, subject to a maximum weekly purchase cap.

Confirmed customer orders occupy the highest priority in the objective function. They are modelled as a prioritised soft constraint: a cumulative constraint ensures that, by any target week, the sum of initial stock and planned production is sufficient to cover all confirmed orders up to that point. Any shortfall is penalised with the highest weight in the objective function, making it the most costly violation the model can incur. Predicted demand and stock targets — safety stock and cycle stock — are similarly modelled as soft constraints with progressively lower penalty weights.

To balance solution quality with computational tractability, and to faithfully represent the operational reality of the company, a set of simplifying assumptions was adopted in the model formulation. These are listed below.

1. **Demand is deterministic.** All demand inputs — confirmed orders, predicted demand, and safety stock and cycle stock targets — are treated as known and fixed for the duration of each model run. These values are provided by the demand planning module prior to each weekly execution and are not updated within a run.
2. **Setup times and unit production times are deterministic.** Both parameters are introduced in the web application (WebApp), as described in Section 3.5, and treated as fixed inputs. Variability in operator performance, machine condition, and product-specific complexity is implicitly captured through the historical estimation process.
3. **Setups are sequence-independent.** A setup is activated whenever a product is produced on a given routing in a given week, regardless of what was produced on that routing in the previous week. Sequence-dependent changeover times are not modelled, as the weekly granularity of the model makes detailed sequencing decisions the responsibility of the short-term planning module.
4. **The shift calendar is a fixed input.** The number of available working days per shift per machine per week is defined by the company's operational calendar and provided as a parameter. The model decides how much of that available capacity to utilise, but does not determine the shift schedule itself.
5. **Operators are fully interchangeable within the same production area.** Workers are not assigned to specific machines but to one of four crew groups: lithography, varnishing, stamping, and assembly. All operators within a group are assumed to be equally capable of operating any machine in that group.
6. **The availability of non-steel inputs is modelled up to the earliest possible replenishment date, and assumed unlimited thereafter.** For non-steel raw materials and consumables — such as varnishes and inks — the model tracks stock levels and scheduled arrivals only

up to the week in which a purchase placed at the current moment could first arrive, given the procurement lead time. Beyond that week, availability is assumed to be unlimited, as procurement decisions for these materials fall outside the scope of the model. This ensures that the capacity constraints imposed by current stock levels are respected in the near term, while avoiding the need to model procurement decisions for materials outside the optimisation scope. Only steel coil purchases are explicitly optimised across the full planning horizon.

7. **Backordering is allowed, subject to penalisation.** Unmet demand — whether confirmed orders, predicted demand, or stock target violations — is penalised in the objective function rather than strictly forbidden. This allows the solver to find feasible solutions under tight capacity conditions. Confirmed orders carry the highest penalty weight, making their fulfilment the model's primary priority.
8. **Warehouse capacity is treated as a hard constraint.** The available capacity of each warehouse is adjusted in the data preparation stage to account for stock of products outside the model scope, as described in Section 3.6. The remaining capacity is enforced as a strict upper bound on the total stock held in each warehouse in each week.
9. **The scrap rate is deterministic and estimated per parent product-component pair.** The average quantity of component scrapped per unit of parent product produced is computed from historical production records and incorporated into the BOM consumption factor as a fixed additive term.
10. **The model is scoped exclusively to Factory 01.** Production operations at Factory 02 are treated as external events. Finished products that require further processing at Factory 02 are represented in the model through their last intermediate product produced at Factory 01, with demand signals shifted back in time by the corresponding inter-factory lead time, as described in Section 3.6.
11. **Minimum lot sizes are enforced per product-routing combination.** A binary setup variable ensures that production only occurs in quantities at or above the defined minimum lot size, preventing the model from planning economically meaningless micro-batches.

With these assumptions established, the model captures the core operational constraints and trade-offs that govern medium-term production planning at the company. Together, they define a tractable and faithful representation of the planning environment, allowing the solver to produce meaningful and actionable results within a practical computational budget. The output is a rolling 32-week production plan specifying for each product, routing, and week, whether to produce and in what quantity, alongside procurement decisions for steel coil and the projected evolution of inventory levels across all BOM levels. This plan is intended to serve as a decision-support tool for the planning team, providing a structured, data-driven starting point that can be refined through human judgment before execution.

4.2 Model Formulation

The optimisation model developed in this dissertation is formulated as a Mixed-Integer Linear Programming (MILP) problem. This choice is motivated by the nature of the decisions involved: production quantities and inventory levels are continuous variables, while setup activation decisions are inherently binary. As discussed in Section 2.4, MILP has been consistently identified in the literature as the dominant modelling approach for tactical production planning problems of this type (Díaz-Madroño et al., 2014; Guzman et al., 2022).

The formulation is multi-level and multi-resource. It is multi-level because the production process spans several BOM levels — from steel coil through intermediate lithographed and stamped components to finished metal cans — and the model explicitly tracks material flows and inventory at each level. It is multi-resource because each product can be produced through different routings, each involving one or more machines across the three production areas of Factory 01: lithography — which includes varnishing —, stamping, and assembly. The routing structure is a distinguishing feature of the formulation: rather than indexing production decisions directly by machine, the model indexes them by routing, allowing a single production run to consume capacity across multiple machines in sequence, reflecting the actual flow of materials through the factory.

The input data required to feed the model — including product master data, routings, BOM structure, initial stock levels, demand forecasts, and configurable parameters — is described in detail in Sections 3.5 and 3.6. The following sections present the complete mathematical formulation: Section 4.2.1 defines the sets, subsets, parameters, and decision variables; Section 4.2.2 presents the objective function; and Section 4.2.3 describes the constraints.

4.2.1 Parameters and Decision Variables

The model's mathematical notation is organised across four tables. Parameters capture the operational environment of the company — such as demand forecasts, production times, BOM consumption factors, and capacity limits — and are treated as known, fixed inputs for each model run. Decision variables, on the other hand, represent the quantities that the model optimises: they encode the decisions that the planning team seeks to determine, including production quantities, inventory levels, setup activations, and steel coil purchases. The decision variables include both continuous non-negative variables, for quantities and inventory levels, and binary variables, for setup decisions.

Table 4.1 defines the primary index sets and subsets, including product subsets and derived index sets used to scope constraints to the relevant product types, routings, weeks, and warehouses. Table 4.2 lists all model parameters, and Table 4.3 lists all decision variables.

Table 4.1: Sets and subsets

Set	Description
P	Set of all products
$W = \{1, \dots, 32\}$	Set of all weeks in the planning horizon
M	Set of all machines
R	Set of all routings
A	Set of all warehouses
MG	Set of all macrogroups
$C = \{1, 2, 3, 4\}$	Crew groups: 1 – Varnishing; 2 – Lithography; 3 – Stamping; 4 – Assembly
$P_{finished} \subset P$	Subset of products that complete their production process at Factory 01, including products transferred to Factory 02 for further processing
$P_{boundary} \subset P_{finished}$	Subset of intermediate products at Factory 01 that serve as proxies for finished products completed at Factory 02
$P_{06} \subset P_{finished}$	Subset of fully assembled finished metal cans (Type 06)
$P_{03} \subset P$	Subset of lithographed sheet products (Type 03)
$P_{intermediate} \subset P$	Subset of intermediate products not including $P_{boundary}$
$P_{coil} \subset P$	Subset of steel coil products
$P_{other} \subset P$	Subset of other purchased input components
$G(p)$	Subset of optional component groups for product p
$C(p, g) \subset P$	Subset of components belonging to group g for product p
$Services(p_{03}) \subset P_{06}$	Subset of Type 06 products served by Type 03 product p_{03}
$p_{03}(p) \in P_{03}$	The Type 03 product associated with Type 06 product $p \in P_{06}$
$P(mg) \subset P$	Subset of products belonging to macrogroup mg
$P_a \subset P$	Subset of products assigned to warehouse a
$R_p \subset R$	Set of valid routings for product p
$R_{out} \subset R$	Subset of outsourced routings
$M_r \subset M$	Subset of machines belonging to routing r
$c(m) \in C$	Crew group to which machine m belongs
$M_c \subset M$	Subset of machines belonging to crew group $c \in C$
$MW \subseteq M \times W$	Subset of active machine-week combinations
$PRW \subseteq P \times R \times W$	Subset of active product-routing-week combinations
$PW \subseteq P \times W$	Subset of active product-week combinations
$PW_{cap} \subseteq P \times W$	Subset of product-week combinations subject to a stock cap constraint

The primary index sets define the domains over which the model's parameters and variables are indexed. Products, defined in P are grouped into macrogroups, collected in set MG : each macrogroup aggregates all optional components that are interchangeable across a set of subsequent products in the BOM. The routing set R defines the distinct sequences of operations — each involving one or more machines in M — through which a product can be manufactured.

Table 4.2: Parameters

Parameter	Definition
$ConfirmedOrders_{p,w}$	Confirmed order quantity of product p for week w
$D_{p,w}$	Demand of product p in week w
$StockStart_p$	Initial inventory of product p at week 1
$StockArrivals_{p,w}$	Exogenous stock arrivals of product p in week w
$SafetyStock_{p,w}$	Safety stock target for product p in week w
$CycleStock_{p,w}$	Cycle stock target for product p in week w
$CycleStock_{mg,w}^{macro}$	Cycle stock target for macrogroup mg in week w
$Cap_{p,w}$	Maximum allowed stock of product p in week w
$MinLotSize_p$	Minimum production lot size for product p
$CoilPurchase_{p,w}^{max}$	Maximum quantity of coil product p that can arrive in week w
$CostPurchase_{p,w}$	Unit purchase cost of product p in week w
$NextPurchaseArrivalWeek_p$	Earliest week in which a purchase of product p placed now could arrive
$ConsumptionFactor_{p_2,p_1}$	Quantity of component p_2 required to produce one unit of product p_1
$Multiplier_p$	Number of units of product $p \in P_{06}$ obtainable from one unit of Type 03 product $p \in P_{03}$
η_{p_2,p_1}	Average scrap of component p_2 per unit of product p_1 produced
$UnitProductionTime_{p,r,m}$	Unit production time (hours) of product p in routing r on machine m
$AvgSetupTime_{p,r,m}$	Average setup time (hours) for product p in routing r on machine m
$WarehouseCapacity_a$	Storage capacity of warehouse a
$NetWeight_p$	Net weight per unit of product p
$Crew_m$	Number of workers required to operate machine m
$TotalCrewShift1_{w,c}$	Total available workers of crew group c for shift 1 in week w
$TotalCrewShift2_{w,c}$	Total available workers of crew group c for shift 2 in week w

Parameter	Definition
$TotalCrewShift3_{w,c}$	Total available workers of crew group c for shift 3 in week w
$PreventiveMaintenance1_{m,w}$	Time consumed by preventive maintenance on machine m in shift 1 of week w , expressed in days
$PreventiveMaintenance2_{m,w}$	Time consumed by preventive maintenance on machine m in shift 2 of week w , expressed in days
$PreventiveMaintenance3_{m,w}$	Time consumed by preventive maintenance on machine m in shift 3 of week w , expressed in days
$WorkingDays1_{w,c}$	Number of working days in shift 1 of week w
$WorkingDays2_{w,c}$	Number of working days in shift 2 of week w
$WorkingDays3_{w,c}$	Number of working days in shift 3 of week w
$Capacity_{r,w}$	Maximum quantity producible via outsourced routing r in week w
$PenalizationConfirmedOrders_p$	Unit penalty cost for unmet confirmed orders of product p
$PenalizationDemand_p$	Unit penalty cost for unmet predicted demand of product p
$PenalizationSafetyStock_p$	Unit penalty cost for safety stock violation of product p
$PenalizationCycleStock_p$	Unit penalty cost for cycle stock violation of product p
$UnitCost_p$	Unit cost of product p
$CostSetup_p$	Setup cost per hour for product p
$CostOutsourcing_{p,r}$	Unit cost of producing product p via outsourced routing r
WACC	Weighted Average Cost of Capital (annualised)
α	Weight of service level costs in the objective function
β	Weight of coil purchase costs in the objective function
γ	Weight of stock holding costs in the objective function
δ	Weight of setup costs in the objective function
ε	Weight of outsourcing costs in the objective function

The parameters define both the product characteristics and the operational environment of the plant. Capacity parameters govern the available production time per machine and the storage limits per warehouse. BOM parameters — namely the consumption factors and the per-unit scrap rates η_{p_2,p_1} described in Section 3.6 — determine the material requirements at each level of the production process. Demand patterns, confirmed orders, and stock targets are provided as inputs

by the demand planning module: safety stock ensures the company is prepared to absorb demand variability, while cycle stock guarantees that sufficient inventory is available to cover the following week's production needs. For each product-routing-machine combination, setup times and unit production times characterise the capacity consumption of each production run. Finally, penalty and cost parameters — covering confirmed order shortfalls, demand violations, stock target breaches, inventory holding, setups and outsourced production — drive the objective function and reflect the company's planning priorities.

Table 4.3: Decision variables

Variable	Definition
$x_{p,w,r} \geq 0$	Quantity of product p produced in week w using routing r
$y_{p_1,p_2,w} \geq 0$	Quantity of component p_2 consumed to produce product p_1 in week w
$k_{p,w,r} \in \{0, 1\}$	1 if product p is produced in week w using routing r ; 0 otherwise
$q_{p,w} \geq 0$	Quantity of coil product p purchased and arriving in week w
$ss_{p,w} \geq 0$	Inventory level of product p at the end of week w
$zsp_{p,w} \geq 0$	Shortage relative to demand of product p in week w
$zco_{p,w} \geq 0$	Shortage relative to confirmed orders of product p up to week w
$zss_{p,w} \geq 0$	Shortage relative to safety stock of product p in week w
$zcs_{p,w} \geq 0$	Shortage relative to cycle stock of product p in week w
$zcs_{mg,w}^{macro} \geq 0$	Shortage relative to macrogroup cycle stock of macrogroup mg in week w
$ta_{m,w} \geq 0$	Total production time (hours) available on machine m in week w
$tt1_{m,w} \geq 0$	Number of shift-1 days allocated to production on machine m in week w
$tt2_{m,w} \geq 0$	Number of shift-2 days allocated to production on machine m in week w
$tt3_{m,w} \geq 0$	Number of shift-3 days allocated to production on machine m in week w

The decision variables encode the planning decisions that the model optimises: from production quantities and setup activations, through inventory levels and material flows across BOM levels, to capacity allocation across machines and shifts, and steel coil procurement. The slack variables zsp , zco , zss , zcs , and zcs^{macro} measure shortfalls against each soft constraint target and are penalised in the objective function when non-zero.

4.2.2 Objective Function

The objective of the model is to minimise a weighted sum of five cost components, as expressed in Expression 4.1:

$$\begin{aligned}
\min \quad & \alpha \cdot \left[\sum_{p \in P_{finished}} \sum_{w \in W} PenalizationConfirmedOrders_p \cdot zco_{p,w} \right. \\
& + \sum_{p \in P_{finished}} \sum_{w \in W} PenalizationDemand_p \cdot zsp_{p,w} \\
& + \sum_{p \in P_{06}} \sum_{w \in W} PenalizationSafetyStock_p \cdot zss_{p,w} \\
& + \sum_{p \in P_{06}} \sum_{w \in W} PenalizationCycleStock_p \cdot zcs_{p,w} \\
& \left. + \sum_{mg \in MG} \sum_{w \in W} PenalizationCycleStock_{mg} \cdot zcs_{mg,w}^{macro} \right] \\
& + \beta \cdot \left[\sum_{p \in P_{coil}} \sum_{w \in W} CostPurchase_{p,w} \cdot q_{p,w} \right] \\
& + \gamma \cdot \left[\sum_{(p,w) \in PW} \frac{WACC}{52} \cdot UnitCost_p \cdot \frac{ss_{p,w-1} + ss_{p,w}}{2} \right] \\
& + \delta \cdot \left[\sum_{p \in P} \sum_{r \in R_p} \sum_{m \in M_r} \sum_{w \in W} CostSetup_p \cdot AvgSetupTime_{p,r,m} \cdot k_{p,r,w} \right] \\
& + \varepsilon \cdot \left[\sum_{p \in P} \sum_{r \in R_p \cap R_{out}} \sum_{w \in W} CostOutsourcing_{p,r} \cdot x_{p,w,r} \right]
\end{aligned} \tag{4.1}$$

where α , β , γ , δ and ε are configurable weight parameters that reflect the company's planning priorities .

The first term penalises service level violations across four dimensions: unmet confirmed orders, unsatisfied demand, safety stock shortfalls, and cycle stock shortfalls, at the individual product level and at the macrogroup level. The second term penalises the cost of steel coil purchases. The third term captures inventory holding costs, computed as the product of the weekly capital cost rate $\frac{WACC}{52}$, the unit cost of each product, and the average stock level between consecutive weeks — approximating the weekly financial cost of capital tied up in inventory. The fourth term penalises setup activations, weighted by the setup time and cost per product-routing-machine combination. The fifth term penalises production through outsourced routings, weighted by a unit outsourcing cost specific to each product and outsourced routing.

4.2.3 Constraints

The constraints are organised into three groups. Group A covers inventory and material flow constraints — stock balances, service level requirements, procurement limits, and warehouse

capacity. Group B covers capacity constraints — machine time, workforce, and setup activation. Group C contains the confirmed orders fulfilment constraints.

A1–A2. Stock Balance of Intermediate Products

$$ss_{p,w} = ss_{p,w-1} - \sum_{p_1 \in P} y_{p_1,p,w} + \sum_{r \in R_p} x_{p,w,r} \quad \forall p \in P_{intermediate}, w = 2, \dots, 32 \quad (4.2)$$

$$ss_{p,1} = StockStart_p - \sum_{p_1 \in P} y_{p_1,p,1} + \sum_{r \in R_p} x_{p,1,r} \quad \forall p \in P_{intermediate} \quad (4.3)$$

Constraints A1 (4.2) and A2 (4.3) define the weekly stock balance for intermediate products. At the end of each week, the inventory level equals the previous week's closing stock, minus the quantity consumed as a component in the production of parent products, plus the quantity produced across all valid routings. A2 handles the initial week separately, replacing the previous week's stock with the known starting inventory $StockStart_p$.

A3.1–A3.5. Service Level Constraints

These constraints enforce cycle stock and safety stock targets as soft constraints, allowing shortfalls to be penalised in the objective function rather than strictly forbidden. Three categories of products are subject to service level constraints, each handled differently.

For Type 06 finished products, the stock of the associated Type 03 lithographed sheets contributes to service level coverage: since Type 03 products are already customer-specific, stock held at that stage can be converted into the corresponding Type 06 products and therefore provides real coverage against a finished goods shortfall. Service level targets are enforced both at the individual product level (A3.1–A3.2) and at the group level (A3.3–A3.4): the individual constraints ensure no single Type 06 product is left uncovered, while the group constraints ensure that the shared Type 03 stock is not over-allocated across the Type 06 products it serves. The parameter $Multiplier_p$ converts units of Type 03 product into equivalent Type 06 units.

Finally, A3.5 enforces cycle stock targets at the macrogroup level, aggregating the stock of all optional components within each macrogroup.

A3.1 (4.4) enforces the individual cycle stock target for each Type 06 product:

$$ss_{p,w} + Multiplier_{p_{03}(p)} \cdot ss_{p_{03}(p),w} \geq CycleStock_{p,w} - zcs_{p,w} \quad \forall p \in P_{06}, \forall w \in W \quad (4.4)$$

A3.2 (4.5) enforces the individual safety stock target for each Type 06 product:

$$\begin{aligned} ss_{p,w} + Multiplier_{p_{03}(p)} \cdot ss_{p_{03}(p),w} - CycleStock_{p,w} + zcs_{p,w} \\ \geq SafetyStock_{p,w} - zss_{p,w} \quad \forall p \in P_{06}, \forall w \in W \end{aligned} \quad (4.5)$$

A3.3 (4.6) enforces the group-level cycle stock target:

$$\sum_{p \in \text{Services}(p_{03})} ss_{p,w} + \text{Multiplier}_{p_{03}} \cdot ss_{p_{03},w} \geq \sum_{p \in \text{Services}(p_{03})} (\text{CycleStock}_{p,w} - zcs_{p,w}) \quad (4.6)$$

$$\forall p_{03} \in P_{03}, \forall w \in W$$

A3.4 (4.7) enforces the group-level safety stock target:

$$\sum_{p \in \text{Services}(p_{03})} ss_{p,w} + \text{Multiplier}_{p_{03}} \cdot ss_{p_{03},w} \geq \sum_{p \in \text{Services}(p_{03})} (\text{CycleStock}_{p,w} - zcs_{p,w} + \text{SafetyStock}_{p,w} - zss_{p,w}) \quad (4.7)$$

$$\forall p_{03} \in P_{03}, \forall w \in W$$

A3.5 (4.8) enforces the cycle stock target at the macrogroup level, aggregating the stock of all optional components within each macrogroup:

$$\sum_{p \in P(mg)} ss_{p,w} \geq \text{CycleStock}_{mg,w}^{\text{macro}} - zcsmg_{mg,w} \quad \forall mg \in MG, \forall w \in W \quad (4.8)$$

In all constraints, the slack variables $zcs_{p,w}$, $zss_{p,w}$, and $zcsmg_{mg,w}^{\text{macro}}$ absorb any shortfall and are penalised in the objective function. This soft formulation allows the solver to find feasible solutions under tight capacity conditions while preserving the priority hierarchy defined in Section 4.1.

A4–A5. Stock Balance of Steel Coil

$$ss_{p,w} = ss_{p,w-1} - \sum_{p_1 \in P} y_{p_1,p,w} + \text{StockArrivals}_{p,w} + q_{p,w} \quad \forall p \in P_{\text{coil}}, w = 2, \dots, 32 \quad (4.9)$$

$$ss_{p,1} = \text{StockStart}_p - \sum_{p_1 \in P} y_{p_1,p,1} + \text{StockArrivals}_{p,1} + q_{p,1} \quad \forall p \in P_{\text{coil}} \quad (4.10)$$

Constraints A4 (4.9) and A5 (4.10) define the weekly stock balance for steel coil. The structure mirrors A1 (4.2) and A2 (4.3), with two additional terms: $\text{StockArrivals}_{p,w}$, which captures exogenous coil deliveries already scheduled at the time of the model run, and $q_{p,w}$, the decision variable representing new coil purchases arriving in week w . The coil stock balance is therefore the only balance constraint where a procurement decision variable appears, reflecting the fact that steel coil is the only raw material whose purchasing is explicitly optimised by the model.

A6. Maximum Weekly Coil Purchase

$$q_{p,w} \leq \text{CoilPurchase}_{p,w}^{\text{max}} \quad \forall p \in P_{\text{coil}}, \forall w \in W \quad (4.11)$$

Constraint A6 (4.11) ensures that steel coil arrivals are restricted to dates permitted by the procurement calendar, so that the production planning model only considers feasible delivery dates.

A7–A8. Stock Balance of Other Purchased Components

$$ss_{p,w} = ss_{p,w-1} - \sum_{p_1 \in P} y_{p_1,p,w} + StockArrivals_{p,w} \quad (4.12)$$

$$\forall p \in P_{other}, w = 2, \dots, NextPurchaseArrivalWeek_p$$

$$ss_{p,1} = StockStart_p - \sum_{p_1 \in P} y_{p_1,p,1} + StockArrivals_{p,1} \quad \forall p \in P_{other} \quad (4.13)$$

Constraints A7 (4.12) and A8 (4.13) define the stock balance for non-steel purchased inputs, such as varnishes and inks. The structure is identical to the coil balance, with the exception that no purchase decision variable appears — procurement of these materials is outside the scope of the model. Crucially, A7 is only enforced up to week $NextPurchaseArrivalWeek_p$, the earliest week in which a purchase placed at the current moment could arrive given the procurement lead time. Beyond that week, availability is assumed unlimited, as described in assumption 6 of Section 4.1.

A9–A10. Stock Balance of Finished Products

$$ss_{p,w} = ss_{p,w-1} - D_{p,w} + \sum_{r \in R_p} x_{p,w,r} + zsp_{p,w} - zsp_{p,w-1} \quad \forall p \in P_{06}, w = 2, \dots, 32 \quad (4.14)$$

$$ss_{p,1} = StockStart_p - D_{p,1} + \sum_{r \in R_p} x_{p,1,r} + zsp_{p,1} \quad \forall p \in P_{06} \quad (4.15)$$

Constraints A9 (4.14) and A10 (4.15) define the weekly stock balance for finished products. Compared to the intermediate product balance, two differences arise. First, demand $D_{p,w}$, which includes confirmed orders, is deducted directly from stock, since finished products are consumed by customer orders rather than by upstream production processes. Second, the slack variable $zsp_{p,w}$ appears as an incremental term: its difference across consecutive weeks captures any demand shortfall that was carried forward from the previous period, ensuring that unmet demand in week $w - 1$ is accounted for in week w rather than lost. This formulation allows the model to represent backlogged demand explicitly, with the accumulated shortfall penalised in the objective function.

A9b–A10b. Stock Balance of Boundary Products

$$ss_{p,w} = ss_{p,w-1} - D_{p,w} + \sum_{r \in R_p} x_{p,w,r} - \sum_{p_1 \in P} y_{p_1,p,w} + zsp_{p,w} - zsp_{p,w-1} \quad (4.16)$$

$$\forall p \in P_{boundary}, w = 2, \dots, 32$$

$$ss_{p,1} = StockStart_p - D_{p,1} + \sum_{r \in R_p} x_{p,1,r} - \sum_{p_1 \in P} y_{p_1,p,1} + zsp_{p,1} \quad \forall p \in P_{boundary} \quad (4.17)$$

Constraints A9b (4.16) and A10b (4.17) define the stock balance for boundary products — intermediate products at Factory 01 that serve both as outputs delivered to Factory 02 and as inputs to other production processes within Factory 01. Compared to the standard finished product balance

A9–A10, an additional consumption term $\sum_{p_1 \in P} y_{p_1, p, w}$ appears, deducting internal consumption from the available stock. This reflects the dual role of boundary products: units produced may either satisfy external demand from Factory 02 or be consumed internally in the continuation of the production process at Factory 01.

A11. Consumed Products per Production

$$\sum_{r \in R_{p_1}} x_{p_1, w, r} = \sum_{p_2 \in C(p_1, g)} y_{p_1, p_2, w} \cdot \left(\frac{1}{\text{ConsumptionFactor}_{p_2, p_1} + \eta_{p_2, p_1}} \right) \quad (4.18)$$

$$\forall p_1 \in P, \forall g \in G(p_1), \forall w \in W$$

Constraint A11 (4.18) links production quantities to component consumption. For each parent product p_1 and each optional component group g , the total quantity produced across all valid routings must equal the total quantity of components consumed within that group, adjusted for both the nominal BOM consumption factor and the per-unit scrap rate. The term $\frac{1}{\text{ConsumptionFactor}_{p_2, p_1}}$ converts the consumed quantity $y_{p_1, p_2, w}$ back into production units, while η_{p_2, p_1} adds the expected scrap loss per unit of p_1 produced. This ensures that the material requirements fed into the model account for expected production losses, as described in Section 3.6.

The optional component group structure means that for products with alternative components, the constraint is enforced at the group level rather than per individual component: the model must consume the correct total quantity from within each group, but is free to choose any combination of components within it.

A12. Warehouse Capacity

$$\sum_{p \in P_a} ss_{p, w} \cdot \text{NetWeight}_p \leq \text{WarehouseCapacity}_a \quad \forall a \in A, \forall w \in W \quad (4.19)$$

Constraint A12 (4.19) ensures that the total stock held in each warehouse does not exceed its available capacity in any week. Stock is measured in terms of net weight, with NetWeight_p converting units of product p into the weight unit used to express warehouse capacity. As described in Section 3.6, $\text{WarehouseCapacity}_a$ reflects only the space available for products represented in the model, having been adjusted in the data preparation stage to deduct the weight of out-of-scope products already occupying each warehouse.

A13. Minimum Lot Size

$$x_{p, w, r} \geq \text{MinLotSize}_p \cdot k_{p, w, r} \quad \forall (p, w, r) \in PRW \quad (4.20)$$

Constraint A13 (4.20) enforces a minimum production lot size for each product-routing-week combination. If a setup is activated for product p on routing r in week w , then the quantity produced must be at least MinLotSize_p . This prevents the model from planning economically meaningless micro-batches that would be infeasible to execute in practice.

A14. Stock Cap

$$ss_{p,w} \leq Cap_{p,w} \quad \forall (p,w) \in PW_{cap} \quad (4.21)$$

Constraint A14 (4.21) enforces an upper bound on the stock level of each product in each week. This constraint is applied selectively to the subset $PW_{cap} \subseteq P \times W$ of product-week combinations for which a stock cap is defined, which corresponds primarily to Make-To-Stock (MTS) products subject to client-authorised production limits. As described in Section 3.6, producing beyond the authorised quantity creates the risk of holding inventory that the client may ultimately not purchase, resulting in obsolete stock. The derivation of $Cap_{p,w}$ is described in Section 3.6.

B1. Production Time per Machine

$$\sum_{\substack{p,r: \\ (p,r,w) \in PRW, r \notin R_{out}, m \in M_r}} (x_{p,w,r} \cdot UnitProductionTime_{p,r,m} + k_{p,w,r} \cdot AvgSetupTime_{p,r,m}) \leq ta_{m,w} \quad \forall (m,w) \in MW \quad (4.22)$$

Constraint B1 (4.22) ensures that the total time consumed on each machine in each week — comprising both production time and setup time across all products and routings that use that machine — does not exceed the total available production time $ta_{m,w}$. The set M_r denotes the machines involved in routing r . Outsourced routings ($r \in R_{out}$) are excluded from this constraint.

B1b. Production Capacity of Outsourced Routings

$$\sum_{p \in P: (p,r,w) \in PRW} x_{p,w,r} \leq Capacity_{r,w} \quad \forall r \in R_{out}, \forall w \in W \quad (4.23)$$

Constraint B1b (4.23) limits the quantity produced through each outsourced routing in each week to a fixed capacity, expressed directly in units rather than time.

B2–B4. Workforce Constraints

$$\sum_{m \in M_c} tt1_{m,w} \cdot Crew_m \leq TotalCrewShift1_{w,c} \quad \forall w \in W, \forall c \in C \quad (4.24)$$

$$\sum_{m \in M_c} tt2_{m,w} \cdot Crew_m \leq TotalCrewShift2_{w,c} \quad \forall w \in W, \forall c \in C \quad (4.25)$$

$$\sum_{m \in M_c} tt3_{m,w} \cdot Crew_m \leq TotalCrewShift3_{w,c} \quad \forall w \in W, \forall c \in C \quad (4.26)$$

Constraints B2 (4.24), B3 (4.25), and B4 (4.26) enforce workforce availability limits for each shift and crew group. For each shift, the total number of workers required across all machines belonging to crew group c — computed as the number of shifts allocated to each machine multiplied by its crew requirement $Crew_m$ — must not exceed the total available workforce for that shift and crew group in that week. The set $M_c \subset M$ denotes the machines belonging to crew group c . Three

separate constraints are defined, one per shift, reflecting the fact that workforce availability may differ across shifts within the same week.

B5–B7. Maximum Number of Shifts per Machine

$$tt1_{m,w} + PreventiveMaintenance1_{m,w} \leq WorkingDays1_{w,c(m)} \quad \forall (m, w) \in MW \quad (4.27)$$

$$tt2_{m,w} + PreventiveMaintenance2_{m,w} \leq WorkingDays2_{w,c(m)} \quad \forall (m, w) \in MW \quad (4.28)$$

$$tt3_{m,w} + PreventiveMaintenance3_{m,w} \leq WorkingDays3_{w,c(m)} \quad \forall (m, w) \in MW \quad (4.29)$$

Constraints B5 (4.27), B6 (4.28), and B7 (4.29) limit the number of productive shifts allocated to each machine in each week for each of the three shifts. The sum of productive shifts $tt_{s,m,w}$ and the days lost to preventive maintenance $PreventiveMaintenance_{s,m,w}$ must not exceed the total number of working days available in that shift, week, and production area $c(m)$, as defined by the company's operational calendar. This ensures that maintenance activities are respected as capacity reductions rather than being overridden by the optimiser in pursuit of higher production volumes.

B8. Total Production Hours per Machine

$$(tt1_{m,w} + tt2_{m,w} + tt3_{m,w}) \cdot 8 = ta_{m,w} \quad \forall (m, w) \in MW \quad (4.30)$$

Constraint B8 (4.30) defines the total production time available on each machine in each week as a function of the productive days allocated across the three shifts. Since each working day corresponds to 8 hours of production time, $ta_{m,w}$ is obtained by multiplying the total productive days across all shifts by 8. This constraint links the shift variables $tt1_{m,w}$, $tt2_{m,w}$, and $tt3_{m,w}$ to the capacity variable $ta_{m,w}$ that appears in B1 (4.22), ensuring consistency between the workforce availability and capacity sides of the model.

B9. Setup Activation

$$\begin{aligned} & x_{p,w,r} \cdot UnitProductionTime_{p,r,m} \\ & \leq 8 \cdot (WorkingDays1_{w,c(m)} + WorkingDays2_{w,c(m)} + WorkingDays3_{w,c(m)}) \cdot k_{p,w,r} \quad (4.31) \\ & \forall (p, w, r) \in PRW, r \notin R_{out}, \forall m \in M_r \end{aligned}$$

Constraint B9 (4.31) links the production quantity variable $x_{p,w,r}$ to the binary setup variable $k_{p,w,r}$. If no setup is activated for product p on routing r in week w — that is, $k_{p,w,r} = 0$ — then the right-hand side collapses to zero and no production is possible. If a setup is activated, the constraint is relaxed by a physically meaningful upper bound: the maximum production time available on machine m in week w , computed as 8 hours per working day multiplied by the total working days across all three shifts for the corresponding production area $c(m)$. This formulation avoids the use of an arbitrary Big-M constant, tightening the linear relaxation and improving solver performance.

C1. Confirmed Orders Fulfilment

$$StockStart_p + zco_{p,w^*} + \sum_{w \leq w^*} \sum_{r \in R_p} x_{p,w,r} \geq \sum_{w \leq w^*} ConfirmedOrders_{p,w} \quad \forall p \in P_{06}, \forall w^* \in W \quad (4.32)$$

Constraint C1 (4.32) ensures that confirmed customer orders are fulfilled in a cumulative sense. For each finished product p and each target week w^* , the sum of the initial stock and all production planned up to and including week w^* must be sufficient to cover all confirmed orders due by that week. The cumulative formulation is deliberately chosen over a per-week enforcement: it allows the model to anticipate production in earlier weeks to cover orders in later weeks, reflecting the flexibility available to the planning team in practice. The slack variable zco_{p,w^*} absorbs any cumulative shortfall and is penalised with the highest weight in the objective function, making confirmed order fulfilment the model's primary priority. This soft formulation ensures feasibility under tight capacity conditions while preserving the priority structure described in Section 4.1.

C1b. Confirmed Orders Fulfilment for Boundary Products

$$StockStart_p + zco_{p,w^*} + \sum_{w \leq w^*} \sum_{r \in R_p} x_{p,w,r} - \sum_{w \leq w^*} \sum_{p_1 \in P} y_{p_1,p,w} \geq \sum_{w \leq w^*} ConfirmedOrders_{p,w} \quad (4.33)$$

$$\forall p \in P_{boundary}, \forall w^* \in W$$

Constraint C1b (4.33) adapts the confirmed orders fulfilment logic of C1 (4.32) to boundary products. The key difference is that cumulative internal consumption is deducted from the left-hand side: units of a boundary product consumed internally at Factory 01 are not available to fulfil external confirmed orders from Factory 02, and must therefore be excluded from the cumulative supply available to cover those orders

4.3 Model Summary

The objective function and constraints defined in Sections 4.2.2 and 4.2.3, constitute the complete MILP formulation of the medium-term production planning model. Collectively, they enforce material flow consistency across all BOM levels, respect finite production capacity and workforce availability, ensure service level targets are pursued with the appropriate priority hierarchy, and penalise operational costs in a configurable and transparent manner.

The model directly addresses the planning challenges identified in Chapter 3: it provides a structured, data-driven basis for production and procurement decisions that replaces the current empirical approach, explicitly balances the trade-off between setup costs and inventory holding costs, and delivers forward visibility over capacity utilisation and stock evolution across the full 32-week horizon. Its outputs serve as inputs to two downstream modules — the coil purchase optimisation module and the short-term production planning module — ensuring coherence across all planning levels. The results obtained from running this model, and their presentation through a dedicated WebApp, are discussed in Chapter 5.

Chapter 5

Results

A model is only useful if the outputs it produces are made available to decision-makers who must analyse and act on them. Without that, the impact of an optimal solution on the company's operations would be limited. It is therefore important to provide decision-makers with appropriate tools to support their day-to-day activities.

This chapter presents the results of the production planning module developed in this dissertation. Section 5.1 characterizes the current planning performance through an analysis of historical production records, establishing a baseline against which the optimised plans are assessed. Section 5.2 describes the postprocessing of the model outputs and the visualizations available in the web application (WebApp). Finally, Section 5.3 reports the computational performance of the model.

5.1 Quality Analysis of the Developed Model

To assess the quality of the plans produced by the model, a set of operational metrics was computed from historical production records and compared against the corresponding values obtained by applying the optimised model to the same historical demand. Using identical input data for both scenarios ensures a fair comparison. This comparison establishes a quantitative baseline against which the gains introduced by the model can be evaluated.

The model was applied to data resembling the historical plan; however, it was not designed to replicate the historical production orders one by one. Instead, the objective was to reproduce the production of finished products and disaggregate these quantities through the Bill of Materials to determine the required production of each intermediate product. This approach preserved the structure of the model while subjecting the historical plan to all model constraints. Fully replicating the historical plan would have required altering several constraints — including BOM consumption with scrap, stock balance, and others — which would have compromised the integrity of the model formulation. To ensure a fair comparison under this approach, the resulting production quantities were verified to be of the same order of magnitude as the historical values, enabling a meaningful comparison between scenarios.

One of the most significant improvements concerns Overall Equipment Effectiveness (OEE), which increased by nearly eight percentage points relative to the as-is scenario. As detailed in Section 5.1.1, this gain is primarily driven by a reduction in the number of setups, which directly improves the availability component of OEE.

A more detailed analysis shows that this improvement is broad-based, occurring across all production areas, but not uniform in magnitude. As shown in Figure 5.1, some areas exhibit considerably greater gains than others, suggesting that the as-is scenario was already closer to efficient operation in certain areas than in others. The remainder of this section examines the role of setups in driving these gains and explains why their impact varies across production areas.

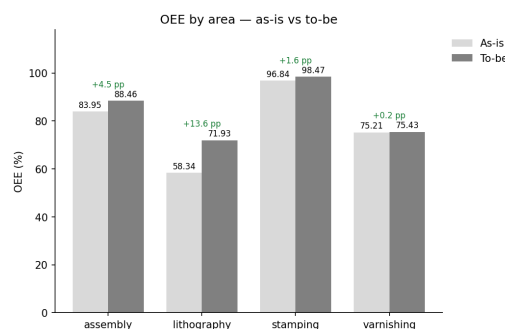


Figure 5.1: OEE by area — As-Is vs. Optimised

The lithography area warrants particular attention. As described in Chapter 3, it is at this stage that products become customer-specific, making it the first point in the production process where SKU differentiation occurs. The resulting variety of products competing for shared lithography resources makes the joint optimisation of lot sizes and setup sequences considerably more complex than in upstream areas, and therefore more dependent on a structured planning approach. Combined with the relatively long setup durations characteristic of this area, the gains achieved by the model in both lot consolidation and setup reduction translate into the most pronounced OEE improvement observed across the four production areas.

5.1.1 Setups Analysis

OEE is composed of three components: availability, performance, and quality. The optimisation model primarily improves the availability component, since reducing the number of setups and increasing lot sizes both directly increase the proportion of available time devoted to production.

Historical records indicate an average of approximately 794 setups per week, corresponding to about 2.17 production orders per product. Thus, in weeks in which a product is produced, it is typically produced more than twice. The optimised plans reduce this to approximately one setup per product per week, consolidating multiple production needs into a single production order.

This average, however, masks considerable variation across the product portfolio. Figure 5.2 presents the distribution of average weekly setups per product for both scenarios. In the as-is scenario, most products require between one and three setups per week, but a substantial number

of outliers exceed this range considerably. In the optimised plans, this distribution is markedly tighter, with the vast majority of products requiring close to a single setup per week and virtually no high-frequency outliers. This indicates that the model's improvement is not limited to the average product, but extends consistently across the full portfolio, including those products that were historically the most setup-intensive.

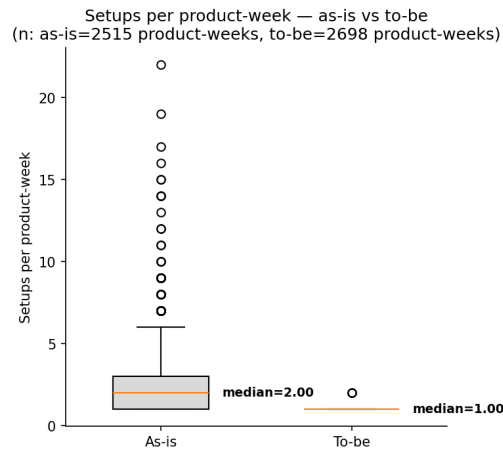


Figure 5.2: Distribution of average setups per product per week — As-Is vs. Optimised

Consistent with the global result, the model also reduces the number of setups within every individual production area, as shown in Figure 5.3.

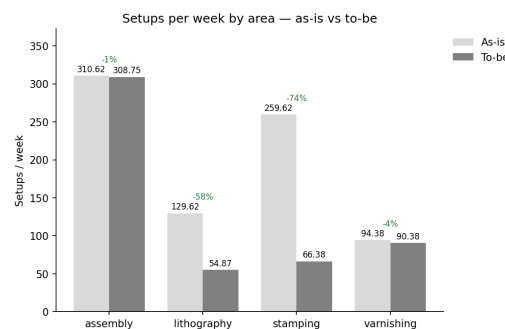


Figure 5.3: Setups by area — As-Is vs. Optimised

Despite this consistent reduction in setup count across all areas, its impact on OEE is not uniform, since the contribution of a setup reduction to availability depends on the duration of each setup relative to total production time. In the stamping area, although the model substantially reduced the number of setups, the effect on OEE was comparatively modest: historical data shows an average setup duration of approximately three minutes in this area, meaning setups account for only a small share of total production time regardless of how often they occur. In lithography, by contrast, the impact was considerably more pronounced — the average setup duration is approximately one hour, so reducing the frequency of setups in this area translates directly into a substantial recovery of available production time, explaining the greatest OEE increase among the four areas.

Taken together, these results show that the OEE improvement cannot be attributed to setup reduction in isolation; it must be interpreted alongside the setup time profile of each production area. Setups had a substantial impact through two combined mechanisms: a reduction in their number, and the reallocation of products to more efficient routings. As a result, products tend to be produced on machines requiring shorter setup times than those historically used, and production cadence is similarly improved, freeing up machine time for additional production orders.

This combined effect is reflected directly in total setup time: average weekly setup hours decrease from approximately 190 to 103, freeing up approximately 87 hours of plant capacity per week that would otherwise be consumed by changeovers. Given that setup operations represent a substantial share of non-productive time in this industry, this reduction represents one of the clearest quantifiable benefits of the optimised planning approach. Figure 5.4 further illustrates this impact by comparing the share of total setup time relative to total production time between the two scenarios.

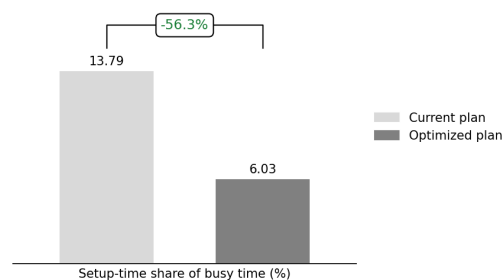


Figure 5.4: Setup-time share of busy time — As-Is vs. Optimised

Examining the allocation of specific products to resources further illustrates the model's decision logic:

- Product 050193 was historically produced entirely on machine PR-56. The model reassigns it to machine PR-34. Although PR-34 requires a longer setup time, its higher production cadence more than compensates for this, resulting in a net gain in available time.
- Product 066951 was historically produced entirely on machine LINH07. The model reassigns it to machine LINH03. Although the production cadence is the same on both machines, LINH03 requires a 6% shorter setup time, resulting in a small but consistent gain.
- Product 036078 was historically produced on a routing involving three machines — two bicolor printers and one varnishing machine — whereas the model reassigns it to a quadricolor printer, which performs varnishing as an integrated step. By consolidating what were previously two separate production stages into one, both total production time and setup time are significantly reduced.

These reassignments illustrate that improving overall plant utilisation sometimes requires moving a product away from its historically preferred machine and onto one where, in isolation, it

performs less efficiently. This is a trade-off the model evaluates at every run, balancing the needs of individual products against the capacity freed for others. Product 03A383 illustrates this trade-off directly: the model reassigns it from the quadricolor printer to a routing using two bicolor printers, freeing quadricolor capacity that is instead allocated to product 036078, described above. Although producing 03A383 on the quadricolor printer would be more efficient in isolation, that routing also requires an additional varnishing step on machine ENVER3, which reduces its net advantage and makes the reallocation worthwhile from the perspective of the plant as a whole.

5.1.2 Lot Size Analysis

The reduction in setup frequency is further reflected the average lot size per production order.

Figure 5.5 presents the average lot size for both scenarios. The optimised plans increase the average lot size by 146.2%, from approximately 5,187 to 12,770 units per production run. This consolidation of production into fewer, larger batches is the direct counterpart of the setup reduction discussed in Section 5.1.1: by producing each product less frequently, the model is able to produce substantially more per run, further reinforcing the availability gains already observed.

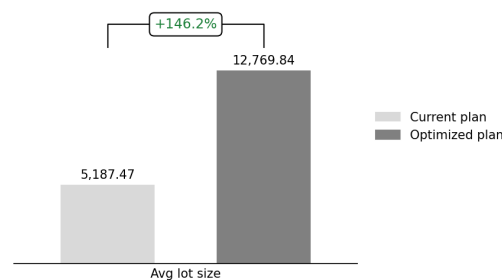


Figure 5.5: Average lot size — As-Is vs. Optimised

Figure 5.6 disaggregates this result by production area. The increase in average lot size is observed in all four areas, although its magnitude varies considerably. The stamping area exhibits both the largest absolute lot sizes and the most pronounced relative increase (+187%), while the varnishing area shows comparatively modest growth (+11%). This variation suggests that the extent to which lot sizes can be increased depends on area-specific characteristics, such as the diversity of products processed and the flexibility of the corresponding routings.

The shift toward larger, more consolidated production runs is also visible in the distribution of lot sizes across all production orders, shown in Figure 5.7. The as-is scenario exhibits a wide distribution including a substantial number of very small lots, reflecting the fragmented production pattern already identified in the setups analysis. In the optimised plans, the distribution is more concentrated around larger lot sizes, and the total number of production lots decreases markedly, from 5,470 to 2,705 — consistent with fewer, larger production runs replacing a greater number of smaller ones.

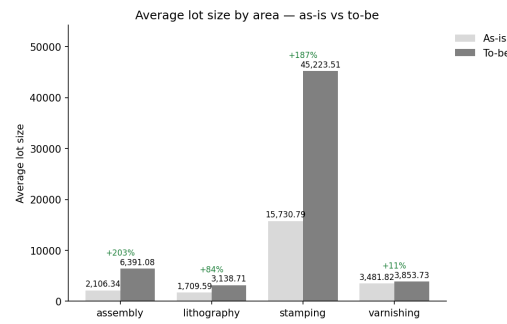


Figure 5.6: Average lot size by production area — As-Is vs. Optimised

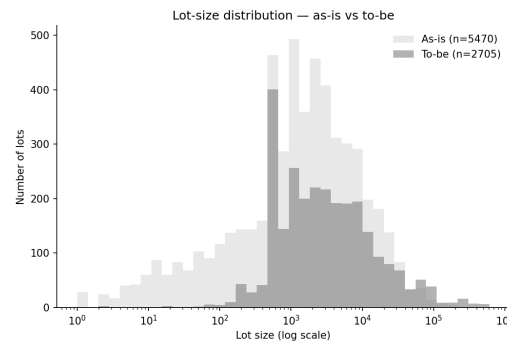


Figure 5.7: Lot size distribution — As-Is vs. Optimised

The effect of this consolidation is particularly visible at the individual product level. Figure 5.8 presents two illustrative examples. For product 055311, the average lot size increases from approximately 18,901 to 208,241 units, while the number of production instances decreases from 87 to 7 over the same period. Product 050149 shows a similar pattern, with the average lot size increasing from approximately 79,424 to 463,264 units, produced only 7 times in the optimised plan. These examples, while limited in sample size, illustrate the underlying mechanism consistently observed across the portfolio: the model trades production frequency for batch size, concentrating the same total demand into fewer, larger production runs.

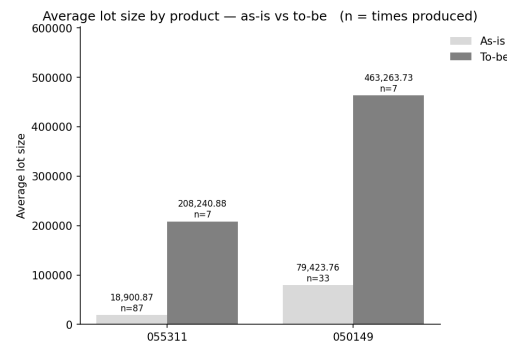


Figure 5.8: Average lot size for selected products

Beyond the direct improvements to production planning, the model developed in this dissertation

serves a broader role within the decision support framework. As described in Chapter 1, the medium-term production plan generated by Module B feeds two downstream modules: the short-term production planning module (Module C) and the coil purchase optimiser (Module D). The quality of those modules' outputs is therefore contingent on the quality of the plan produced here. A more structured and setup-efficient plan translates into more reliable short-term scheduling inputs, while a more accurate coil demand signal — disaggregated by product and week — enables the coil purchase model to optimise steel procurement under tighter and more realistic constraints. In this sense, the gains demonstrated in this section should be understood not only as improvements to the medium-term plan itself, but as upstream quality improvements that propagate through the entire planning chain.

5.2 Output Preparation and Visualization

The optimisation modules described throughout this dissertation produce results that are ultimately made available to decision-makers through a set of visualization plots in the WebApp. However, the raw solver output — consisting solely of the optimal values of the decision variables — cannot directly feed those visualizations. A postprocessing stage is therefore required to transform and structure those values into the dataframes the application consumes.

Data preparation plays a critical role in the modelling pipeline: only well-structured, consistent input data allows the model to produce reliable outputs, a principle broadly recognized in the data science literature (Kilkenny and Robinson, 2018). The pipeline does not end at the model output, however. Raw results must be processed and organized before they can support the range of analyses and visualizations required by decision-makers. In the context of this work, both stages were addressed: input data preparation was covered in Chapter 3, and the postprocessing of the model outputs is described in the present section.

Eight output tables were prepared from the results of the model, complemented by selected input parameters, where needed, to support the intended analyses. Throughout this section, the preparation of each table and the visualizations it feeds are addressed jointly, following the structure of the WebApp.

The WebApp opens with a summary panel displaying a set of global Key Performance Indicators (KPIs) cards for the production planning module, as shown in Figure 5.9. These cards present the following metrics: Total Capacity, defined as the average utilisation rate of production resources over the next four weeks; Planned Orders for the next four weeks; average Stock Coverage in days; average Safety Stock Coverage in days; number of Orders Delayed in the next week; global OEE for the next four weeks; and On-Time In-Full (OTIF) for the next four weeks. Each card also displays the variation of the respective metric relative to the previous model run, allowing the planning team to assess week-on-week changes in the solution quality at a glance.

The chart in Figure 5.10 presents the utilisation rate across the four production areas — varnishing, lithography, stamping, and assembly — for each of the first four planning weeks.

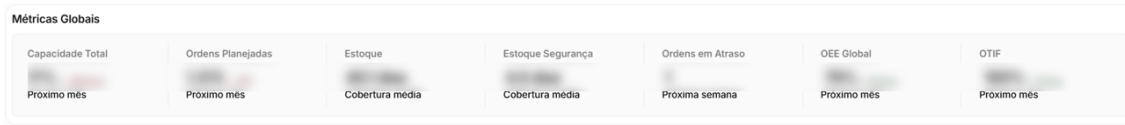


Figure 5.9: Global metrics cards

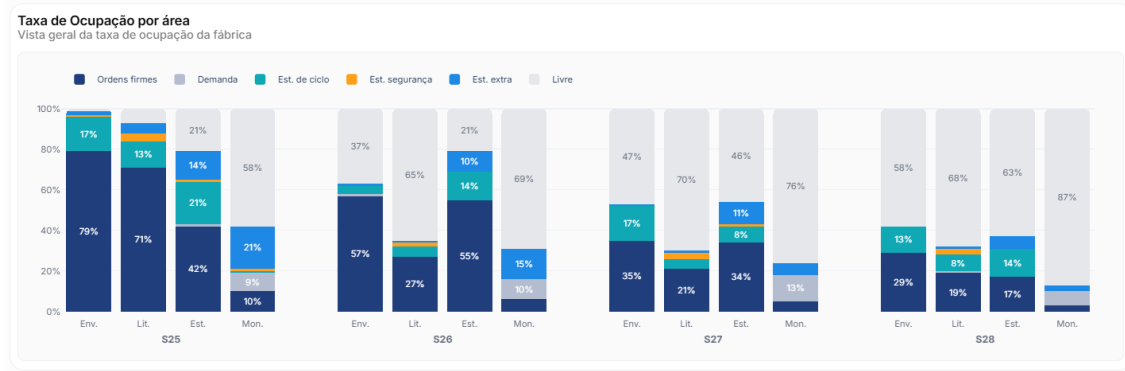


Figure 5.10: Capacity utilization

Each bar is decomposed into the capacity allocated to firm orders, forecast demand, cycle stock, safety stock, and surplus stock, with the remaining portion representing free capacity. This breakdown allows the planning team to assess not only the overall utilisation rate of each area but also the nature of the load driving it.

In addition to the visualizations described above, to provide a more detailed view of stock levels, five cards are presented — one per product area: raw material inputs, varnished sheets, lithographed sheets, components, and final cans — as shown in Appendix A, Figure A.1.

Each card displays the utilisation rate of the resources producing that product type, alongside four stock-level indicators: the number of products with zero stock, the number below cycle stock, the number below safety stock, and the number at a stable stock level.

A weekly stock balance chart is present in Appendix A, Figure A.2. This stacked bar chart presents, for each of the first four planning weeks, the decomposition of stock into cycle stock, safety stock, and surplus stock, alongside the quantities produced, delivered, and delayed. The chart can be filtered by item, customer, or product area, allowing the planning team to drill into specific segments of the plan.

Also, a detailed plan table is available and presented in Appendix A. This table provides a comprehensive view of the full production plan, displaying for each item and week the corresponding production area, assigned resource, planned quantity, setup and production times, opening and closing stock, and the quantities consumed, delivered, and delayed. The table can be filtered by customer, item, area, machine, and date, and can be exported to Excel for further analysis.

A configurations page dedicated to the medium-term production planning module is also available, allowing the user to define the objective function weight parameters α , β , γ , δ , ε prior to each model run. Additional configuration pages, shared across modules but applicable to the

production planning model as well, permit the definition of warehouse capacities, shortage costs, and other parameters; these are also presented in Appendix B.

5.3 Computational Results

The optimisation model formulated in Chapter 4 was implemented in Python using the Pyomo library. The instances were solved using the commercial solver Gurobi, version 11.0.3, on an AMD EPYC 7642 48-Core Processor with 15.4 GB of available RAM, with the limitation of four threads.

Given the multi-objective nature of the model, a two-phase sequential solving strategy was adopted to reflect the hierarchical priority of the objective function components. In the first phase, the model is solved considering only service level costs, ensuring that demand fulfilment is optimised without interference from secondary objectives. In the second phase, the shortage variables are fixed to the values obtained in the first phase, and the model is re-solved — using the first phase solution as a MIP start — optimising the remaining cost components: stock holding, setup, coil purchase, and production outsourcing costs. This decomposition guarantees that service level is not sacrificed in favour of cost efficiency, while still allowing the model to find an efficient solution with respect to the remaining objectives.

The instance described below corresponds to an 8-week planning horizon and reflects the full scale of the problem at Factory 01. Table 5.1 presents the problem dimensions before and after presolve for both phases.

Table 5.1: MIP dimensions before and after presolve

	Phase 1		Phase 2	
	Original	Presolved	Original	Presolved
Rows (constraints)	250,186	140,795	250,186	137,049
Columns (variables)	352,350	211,703	352,350	195,401
Nonzeros	1,137,963	738,951	1,096,014	704,002
Binary variables	48,320	46,362	48,320	46,087
Continuous variables	304,030	165,341	304,030	149,314
Presolve time (s)	8.59		11.71	

The presolve phase significantly reduces the problem size in both cases. In Phase 1, the number of rows is reduced by approximately 42% and the number of columns by approximately 40%. In Phase 2, the reduction is even more pronounced — approximately 45% and 44% respectively — reflecting the additional fixing of shortage variables inherited from Phase 1.

Table 5.2 summarises the computational performance of both phases.

Table 5.2: Computational performance by phase

Phase	Time to first solution (s)	Total solve time (s)	Nodes explored	Final MIP gap (%)
Phase 1	351	4,559.96	3,549	0.99
Phase 2	15	254.98	784	0.32

The two phases exhibit markedly different computational behaviours. Phase 1, solving for service level costs alone, requires significantly more time to find the first feasible solution (351 seconds) and explores a larger portion of the branch-and-bound tree before the gap tolerance is met, terminating at 4,559.96 seconds with a final gap of 0.99%. Phase 2 benefits from the MIP start provided by Phase 1 and from the reduction in problem size resulting from fixing the shortage variables, allowing it to find a first feasible solution in 15 seconds and to terminate in 254.98 seconds with a final gap of 0.32% — well within the imposed tolerance.

It should be noted that this instance corresponds to an 8-week planning horizon, while the full rolling horizon covers 32 weeks. In practice, the model is executed on a weekly basis with a rolling horizon, meaning that the computational time reported here is representative of a single weekly run rather than the full planning horizon.

Computational performance represented a significant challenge throughout the development of this work. The scale of the problem — nearly 250 thousand constraints and over 300 thousand variables before presolve — frequently resulted in convergence difficulties, with the solver failing to find feasible solutions within acceptable time frames. This had a direct impact on the development process, as the ability to run the model and inspect its outputs is essential for debugging the formulation, validating constraints, and iterating on modelling decisions. In practice, many such iterations were rendered infeasible by the solver’s inability to converge in reasonable time.

Several Gurobi parameters were tested in an attempt to improve convergence, including cutting plane strategies and the NoRel heuristic, which bypasses the root relaxation entirely in favour of finding feasible solutions quickly. However, empirical testing indicated that allowing Gurobi to manage these parameters automatically — using default settings — consistently produced the best results for this instance. The two-phase solving strategy described above was also motivated in part by these convergence difficulties, as decomposing the objective function reduced the complexity faced by the solver in each individual phase and improved the reliability of obtaining feasible solutions within the time limit.

Chapter 6

Conclusion

This study had as its main objective the medium-term production planning of a metal packaging factory, over a horizon of 32 weeks — roughly eight months — culminating in an output specifying what products to produce, in which quantities, when, and on which production resources. To this end, a MILP optimisation model was developed within the project’s broader framework.

The project comprised four interconnected modules. Module A handled demand planning, forecasting the coming months’ demand and computing the required safety stock for each finished product. Module B, the core of this dissertation, is the tactical production planning optimisation model: taking the outputs of Module A as inputs and feeding its own outputs to Modules C and D, it occupies a pivotal position within the framework, connecting all three remaining modules. Module C translates these weekly decisions into shift-level production planning, converting the weekly planner into an operational tool. Module D is the steel coil purchase optimiser, crossing the steel requirements derived from Module B with each supplier’s properties to define the optimal purchasing plan.

The approach for the tactical production planning was to develop a Mixed-Integer Linear Programming model, combining continuous and binary variables. The binary variables govern setup activation, which in turn conditions the associated production quantities; all remaining variables are continuous.

The production reality in this industry involves a multi-level bill-of-materials spanning six product-type levels, from raw materials to finished goods. Products can be manufactured on multiple resources, each with its own assigned production and setup times. The model itself combines a large set of constraints that mirror this production reality with a single objective function that minimises the total relevant cost — comprising service-level penalties, inventory holding, setup, and steel purchase costs. Its outputs are presented through a WebApp, fed by tables prepared within this study, which displays the weekly plan through a range of visualisation charts and lets users configure the model’s parameters, making the results accessible to decision-makers.

Meaningful results depend critically on the quality of the input data. In this setting, the data originated from an Enterprise Resource Planning (ERP) system that stores target values rather than actual production figures, which required an extensive cleaning and preparation effort before the

model could be run. A central lesson from this work is therefore that the quality of any tactical plan produced for this factory is bounded by the quality of the underlying data — a constraint that shaped both the preprocessing pipeline and the confidence that can be placed in the results.

The developed model focused exclusively on Factory 01, although the company for which this project was carried out operates two active factories. This introduced a planning difficulty: for certain finished products whose demand was predicted by Module A, part of the production process falls outside the scope of Factory 01. A dedicated data treatment was therefore required to translate the demand for these finished products into the corresponding intermediate-product requirements that Factory 01 must produce — the boundary products subsequently handed off to Factory 02.

A deliberate aim throughout was to keep the model dynamic. The addition of new machines or products is accommodated natively, provided they are correctly inserted into the company's ERP system, and the WebApp plays an important role in this logic, as many of the model's inputs are drawn from it. The model is also prepared to handle new additions such as outsourced machines mimicking in-house machines for selected groups of products. Since businesses are always changing and this company may grow in the future, expanding both its in-house and outsourced operations, a model prepared to absorb such expansions without requiring intervention from its developers is a valuable asset.

The model was implemented in Python using the Pyomo library and solved with Gurobi. Its scale, however, proved to be the most significant limitation encountered during the project: with hundreds of thousands of constraints and over 350 thousand variables — of which almost 50 thousand were binary — even for a horizon of only eight weeks, 25% of the full solving horizon, the model struggled to converge to a proven optimal solution within a feasible time. This had a direct practical cost, as every modelling change became difficult to validate: without convergence to optimality in reasonable time, distinguishing the effect of a change from the residual optimality gap was not straightforward. Different configurations of the Gurobi solver were explored and tested in an effort to improve solving time.

Despite this limitation, the model translated into measurable improvements for the factory's planning process, concentrated on two axes. First, the average number of setups per product produced per week was reduced by more than 50%, as the model consolidated production into larger, less fragmented lots and avoided unnecessary changeovers — releasing capacity that was previously lost to setup time. Second, this gain in setup efficiency contributed to an improvement of almost 10% in Overall Equipment Effectiveness (OEE), reflecting a more productive use of the available equipment across the planning horizon.

A tactical production planning module can have, as demonstrated throughout this dissertation, a significant impact on a company — and, beyond that, on the entire supply chain. By improving the planning process, companies can strengthen their procurement, preparing for the future with sufficient lead time. Suppliers can thus be informed earlier of their clients' requirements, enabling better service across the whole supply chain. Beyond this, by raising the productivity of existing resources, modernising the planning process, and cutting the material and machine time lost to setups, work of this kind contributes to three of the United Nations' Sustainable Development

Goals, detailed in Appendix C.

Several directions remain open for future work. The most immediate is the completion of Module D, the steel coil purchase optimiser — formulated as a two-stage stochastic program that represents demand uncertainty through a set of scenarios — whose mathematical formulation has already been developed and whose implementation will extend the framework presented here. At present, Modules B and D operate sequentially: Module B determines the production plan and passes the resulting steel requirements to Module D, so the two are coordinated but not jointly optimised. Optimising them together represents the most ambitious direction for future work, as it would allow purchasing and production decisions to be balanced against one another rather than resolved in sequence. A further natural extension is to broaden the model's scope beyond Factory 01 to encompass Factory 02, removing the need for the boundary-product treatment described above and enabling integrated planning across both sites.

Situated at the centre of the four-module framework, Module B converts the demand and safety-stock figures from Module A into the production decisions that drive Modules C and D, making it the connective core on which the wider planning solution rests. By bringing the company's tactical production planning into a single optimisation model — tailored to a demanding reality of fully customised products, multi-stage manufacturing, and a deep bill-of-materials — this dissertation replaces a process previously guided by separate, empirical rules with a structured and transparent basis for weekly decisions. Its value is ultimately bounded by the quality of the data that feeds it, but, once the framework is fully deployed, it offers the factory's planning team a solid foundation for medium-term production planning and a starting point from which the remaining modules can be completed.

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Appendix A

Other WebApp Visualizations

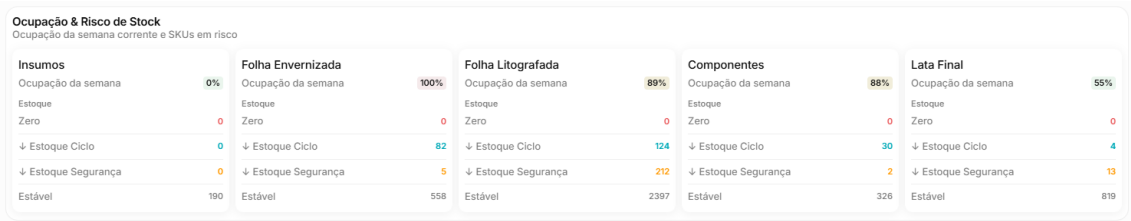


Figure A.1: Stock levels cards



Figure A.2: Stock balance



Figure A.3: Crew usage

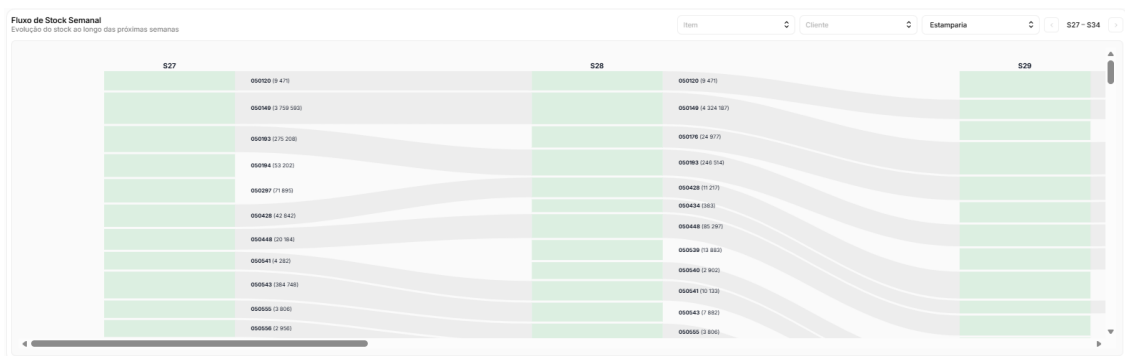


Figure A.4: Stock flow

Plano Detalhado												
Artigo	Área	Máquina	Semana	Stock Inicial	Quantidade Produzida	Tempo Produção	Tempo Setup	Quantidade Consumida	Quantidade Entregue	Quantidade Atrasada	Stock Final	
Estamparia	PR-21	27-2026	28522	0	0.00h	0.00h	6521	0	0	22001		
	PR-53	27-2026	10229	0	0.00h	0.00h	2340	0	0	7889		
	SOLDA-18L	27-2026	10229	0	0.00h	0.00h	2340	0	0	7889		
	US-51	27-2026	10229	0	0.00h	0.00h	2340	0	0	7889		
	PR-53	27-2026	2181	6521	11.4th	-	5532	0	0	3171		
	SOLDA-18L	27-2026	2181	0	0.00h	0.00h	5532	0	0	3171		
	US-51	27-2026	2181	6521	11.4th	-	5532	0	0	3171		
	Envernizamento	ENVER3	27-2026	0	0	0.00h	0.00h	0	0	0	0	
	Envernizamento	ENVER3	27-2026	0	0	0.00h	0.00h	0	0	0	0	
Envernizamento	ENVER3	27-2026	214	0	0.00h	0.00h	214	0	0	0		
	ENVER3	27-2026	0	0	0.00h	0.00h	0	0	0	0		
	ENVER3	27-2026	8430	0	0.00h	0.00h	8160	0	0	1270		
	ENVER3	27-2026	581	3697	0.51h	0.50h	2278	0	0	0		
	Estamparia	PR-30	27-2026	406160	446448	30.95h	-	486832	0	0	365776	
	Envernizamento	ENVER3	27-2026	514	0	0.00h	0.00h	514	0	0	0	
	Envernizamento	ENVER3	27-2026	0	0	0.00h	0.00h	0	0	0	0	
	Envernizamento	ENVER3	27-2026	0	0	0.00h	0.00h	0	0	0	0	
	Envernizamento	ENVER3	27-2026	30819	0	0.00h	0.00h	5948	0	0	24870	
Envernizamento	ENVER3	27-2026	17353	0	0.00h	0.00h	9312	0	0	8041		
	ENVER3	27-2026	0	0	0.00h	0.00h	0	0	0	0		

Figure A.5: Detailed plan table

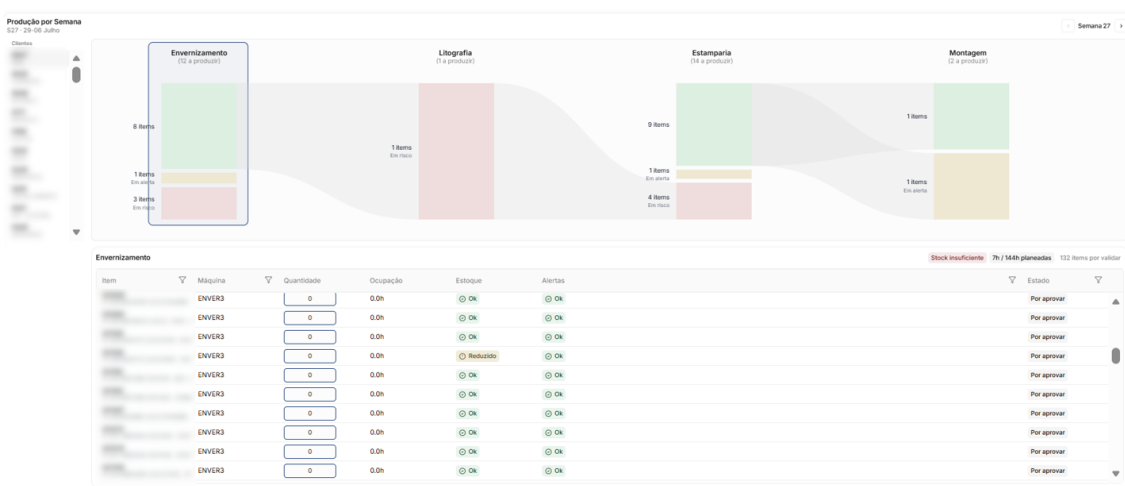


Figure A.6: Plan validation table

Appendix B

Configurations Pages



Figure B.1: Weight parameters selection

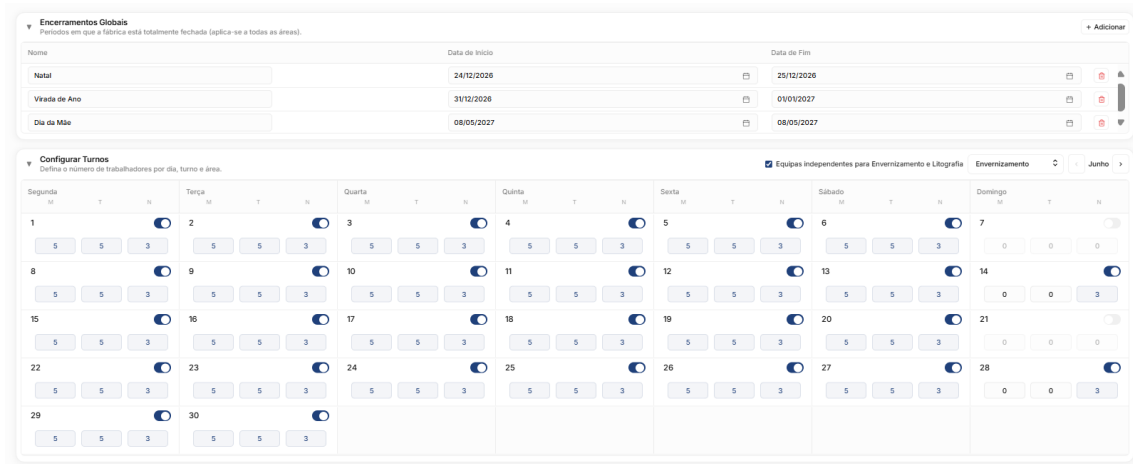


Figure B.2: Working days calendar

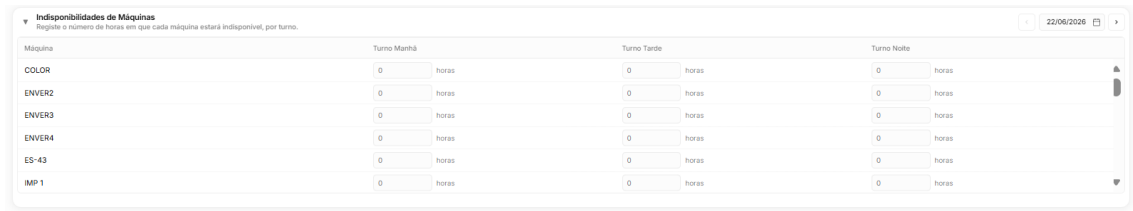


Figure B.3: Machines unavailability

▼

Tempo de Entrega ao Cliente

Parametrize o lead time de entrega por cliente, com valor default e exceções.

Lead Time (dias):

2

▼

Exceções por cliente

Cliente

Lead Time (dias)

Selecionar cliente

5

+ Adicionar

▼ WACC

Defina o custo médio ponderado de capital para cálculos financeiros.

WACC (%):

Custo Setup

Parametrize o custo de setup por máquina.

Envenizamento ↕

Máquina	▼	Valor Reais/Tempo	Valor Reais/Unidade
ENVER3			
ENVER4			
QUAD			

Custo Atraso e Incumprimento de Estoque de Ciclo

Parametrize o custo do atraso de uma encomenda e incumprimento de estoque de ciclo.

Custos Default

Incumprimento Ordens Confirmadas

Incumprimento Stock de Ciclo

Incumprimento Demanda Prevista

Incumprimento Stock de Segurança

Custos por Cliente

Adicionar ➕

Cliente	▼	Inc. Ordens Confirmadas	Inc. Demanda Prevista	Inc. Stock de Ciclo	Inc. Stock de Segurança
No Rows To Show					

[illegible]

▼

Lead Times de Insumos

Defina o lead time padrão para os Insumos.

Lead Time Default (dias):

5

▼

Lead Time por Tipo de Insumo

Defina o lead time por tipo de insumo.

Tipo de Insumo

Veriz

Tinta

Lead Time (dias)

5

5

Capacidade de Armazém

Defina a capacidade de cada armazém.

Armazém	Capacidade	Unidade
Armazém 10	2000	Toneladas
Armazém 15	45	Toneladas
Armazém 20	400	Toneladas
Armazém 21	100	Toneladas
Armazém 22	100	Toneladas
Armazém AL	120	Toneladas
Armazém 25	100	Toneladas
Armazém 30	600	Toneladas
Armazém 35	450	Toneladas
Armazém 36	400	Toneladas
Armazém 23	80	Toneladas
Armazém 40	250	Toneladas
Armazém 45	30	Toneladas
Armazém 50	200	Toneladas
Armazém 08	50	Toneladas
Armazém 14	120	Toneladas
Armazém 19	100	Toneladas
Armazém AN	60	Toneladas

Figure B.8: Warehouses capacities

Cadências

Parametrize as cadências de cada máquina por tipo de produto.

Grupo de Produto

Descrição

Cadência (un/h)

61	18 LITROS HOMOLOGADA	1428.571
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Setups

Configuração dos tempos de setup por tipo de setup e máquina.

Área	Máquina	Tipo de Setup	Tempo de Setup (min)
Envernizamento	ENVER3	Formato	30.0
Envernizamento	ENVER3	Mudança de Código de Produto	10.0
Envernizamento	ENVER3	Grupo de Verniz	20.0
Envernizamento	ENVER3	Produto de Verniz	20.0
Envernizamento	ENVER4	Formato	30.0
Envernizamento	ENVER4	Grupo de Verniz	20.0
Envernizamento	ENVER4	Produto de Verniz	20.0
Litografia	IMP 1	Formato	30.0
Litografia	IMP 1	Produto de Tinta	10.0

Figure B.9: Cadences and setups

Terceirização de Produção

Adicione e parametrize informação de terceirização.

Máquina Cajamar

Máquina Terceirização

Grupos Produtos

Capacidade (un/semana)

Custo (R\$/un)

US-S1	SOLDA-05L	• 024002 • 024003	1000	1
US-S1	SOLDA-18L	• 026002 • 026003	1000	1
US-S1	SOLDA-BALDE	• 026201 • 026204 • 026205	1000	1

Figure B.10: Outsourced production

Appendix C

Sustainable Development Goals

Table C.1: Alignment of the dissertation with the United Nations Sustainable Development Goals.

Goal	Target	Explanation
SDG 8 — Decent Work and Economic Growth	8.2 — Achieve higher levels of economic productivity through diversification, technological upgrading and innovation	The proposed medium-term production planning model improves overall equipment effectiveness (OEE) and capacity utilisation, directly contributing to productivity gains at the factory level without requiring additional labour or capital investment.
SDG 9 — Industry, Innovation and Infrastructure	9.4 — Upgrade infrastructure and retrofit industries to make them sustainable, with increased resource-use efficiency	The decision support framework replaces manual spreadsheet-based planning with a MILP optimisation model, modernising the production planning process and enabling more efficient allocation of productive resources across all manufacturing areas.
SDG 12 — Responsible Consumption and Production	12.5 — Substantially reduce waste generation through prevention, reduction, recycling and reuse	By optimising production lot sizes and reducing the number of setups, the model decreases the amount of raw material and machine time wasted during production changeovers, contributing to leaner and more resource-efficient manufacturing operations.